

S.A ENGINEERING COLLEGE, CHENNAI - 77

(An Autonomous Institution Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

2021-2025 BATCH

Sl.No

NAME

MAPPING

COURSE OUTCOMES

Year:2021-2022(ODD) SEMESTER I

HS1101A - Technical English

C101.1	k2	Describe the memory of the previously learnt grammatical concepts and apply them in technical contexts.
C101.2	k2	Interpret the professional proficiency by preparing the formal correspondence with global standards.
C101.3	K2	Explain information and interrelate the technical and general texts.
C101.4	K3	Use verbal and technical information concisely with coherence and cohesion.
C101.5	K3	Show appropriate statements representing the global situations.

1

C102.1	k2	Describe the basic concepts of differentiation rules to find the maxima and minima for function of one variable.
C102.2	K3	Apply the knowledge of basic application problems involved in first order differential equation.
C102.3	k3	Determine the solution of second order linear differential equations with constant coefficients.
C102.4	k3	Show the concept of ordinary derivatives into partial derivatives and apply them to find extreme values of the functions of two variables and series approximation of the function of two variables.
C102.5	k3	Calculate the double and Triple integration techniques over a region in two dimensional and three-dimensional geometry.

2

PH1101A - Applied Physics

C103.1	K2	Explain the basics of properties of matter and its applications
C103.2	K2	Describe the concepts of waves and optical devices and their applications in Laser and fiber optics
C103.3	K3	Use the properties of thermal materials and its applications
C103.4	K3	Apply the advanced physics concepts of quantum theory and its application in one dimensional box
C103.5	K3	Apply the different types of crystals structures and different crystal growth techniques.

3

CY1101A- Engineering Chemistry

C104.1	K2	Explain the hardness of water, related problems and its treatment.
C104.2	K3	Apply phase rule to construct phase diagram and predict the low melting alloys.
C104.3	K3	Illustrate conventional and non-conventional energy sources and basics of electrochemistry, Types of batteries, their reactions and the importance.
C104.4	K3	Demonstrate the use of fuels in real-world applications by analyzing their calorific values.
C104.5	K4	Analyze the basics, types, preparation methods and recent trends in nanomaterial.

4

CS1101A - Problem Solving and Python Programming

C105.1	K2	Discuss the basics of algorithmic problem solving.
C105.2	K3	Illustrate Python programs with data expressions and statements.

5	C105.1	K1	Apply control flow and python functions for problem solving.
	C105.6	K1	Apply data structures such as lists, tuples, and dictionaries in Python for problem solving.
	C105.8	K1	Examine file concepts to read and write data from/to files.
ME1101A - Engineering Graphics			
6	C106.1	K1	(Describe existing national standards of engineering drawing and visualization concepts)
	C106.2	K2	Explain the projection of Points, Lines and Plane surfaces
	C106.3	K2	Summarize the projection of Solids
	C106.4	K2	Interpret the section of solids
	C106.5	K2	Discuss the development of surfaces
BS1101A - Physics and Chemistry Laboratory			
7	C107.1	K2	Describe fundamental concepts of mechanical properties analysis
	C107.2	K2	Explain the fundamental principles of acoustic properties, and learn about semiconductor properties
	C107.3	K2	Summarize the fundamental principles of optical properties analysis.
	C107.4	K3	Apply hands-on knowledge in the quantitative chemical analysis of water.
	C107.5	K3	Solve the basics of instrumental analysis-conductivity meter, and potentiometer
CS1102A - Problem Solving and Python Programming Laboratory			
8	C108.1	K2	Interpret the behavior of programs that use decision-making and iteration.
	C108.2	K3	Demonstrate the functions to solve mathematical problems
	C108.3	K3	Use strings for structuring Python programs.
	C108.4	K3	Illustrate compound data using Python lists, tuples, and dictionaries.
	C108.5	K3	Demonstrate file concepts to write programs to read and write data.
Year:2021-2022(EVEN) SEMESTER II			
HS1201A- English for Communication			
9	C109.1	K1	Identify technical and general texts using appropriate diction.
	C109.2	K1	Recall various styles of drafts in formal and informal contents.
	C109.3	K3	Use the verbal and technical skills in the given technical contexts.
	C109.4	K3	Apply meaningful utterances to present intense technical knowledge.
	C109.5	K4	Interpret key ideas by reviewing lexicon across the language in syntactical contexts.
MA1201A - Complex Variables and Transforms			
10	C110.1	K2	Explain the concept of divergence and curl to Calculate the line, surface and volume integrals using vector integration.
	C110.2	K3	Solve the problems under analytic functions and construction of analytic function using C-R equation.
	C110.3	K4	Classify the singularities and pole, find residues and evaluate complex integrals using residue theorem.
	C110.4	K4	Outline the concepts of Laplace Transforms and its properties and to solve the differential equations.
	C110.5	K4	Analyze the properties of Z-transform and solve the difference equation.
PH1201A - Materials Science			
	C111.1	K2	Describe the fundamental principles of conducting materials and analyze their behavior under varying temperature conditions.

11	C111.2	K2	Explain knowledge on semiconductor physics, including carrier concentration, and apply it to the design and analysis of semiconductor devices.
	C111.3	K3	Apply the properties of magnetic and superconducting materials, and apply their concepts in practical engineering systems
	C111.4	K3	Examine knowledge on polarization mechanisms in dielectric materials and its applications
	C111.5	K4	Analyze the properties and applications of advanced nano materials and biomaterials for various engineering applications.
F.E1202A- Basic Electrical, Electronics and Measurement Engineering			
12	C112.1	K2	Summarize key principles of fundamental electrical laws and theorems.
	C112.2	K2	Describe DC & AC static and rotating machines
	C112.3	K2	Explain renewable energy sources, various electrical protective devices
	C112.4	K2	Discuss the fundamentals of electronic circuits
	C112.5	K3	Apply the working principle & operation of measuring instruments and transducers
CS1201A - Programming in C			
13	C113.1	K2	Interpret the basic syntax and semantics of the programming language constructs.
	C113.2	K3	Examine derived data types like arrays in solving problems.
	C113.3	K3	Illustrate problem decomposition and reconstruction through multiple user-defined function methods.
	C113.4	K3	Examine user-defined data types like structures and unions and its applications to solve problems.
	C113.5	K3	Apply file handling techniques to develop programs using sequential and random-access processing
CS1202A - Fundamentals of Computing			
14	C114.1	K2	Interpret the major components of a computer system and architecture.
	C114.2	K3	Illustrate various indexing strategies in different database systems
	C114.3	K2	Discuss the programming languages and Operating System concepts
	C114.4	K3	Use HTML and CSS basics to structure and control website layouts as per requirements
	C114.5	K2	Explain the basic concepts of networking.
CS1203A - Programming in C Laboratory			
15	C115.1	K3	Demonstrate C programs for simple applications making use of basic constructs
	C115.2	K3	Examine C programs for control statements
	C115.3	K3	Illustrate C programs involving arrays, strings and pointers
	C115.4	K3	Demonstrate C programs involving functions, and structures
	C115.5	K3	Apply sequential and random-access file processing techniques to develop suitable programs
GE1201A - Engineering Practices Laboratory			
16	C116.1	K2	Describe the basic principle of welding
	C116.2	K2	Explain the basic principle of Sheet Metal Work
	C116.3	K2	Discuss the basic principle of joints
	C116.4	K2	Summarize the basic principle of Machine assembly practice
	C116.5	K3	Apply the basic principle of Drilling

CS1204A - Fundamentals of Computing Laboratory		
17	C117.1	K2 Demonstrate the features of MS Office for designing documents, presentations, and Excel sheets.
	C117.2	K2 Examine how markup languages are used to design static web pages for specific needs and interests.
	C117.3	K2 Discuss the Networking Concepts.
	C117.4	K3 Use typical data definitions and manipulation commands.
	C117.5	K3 Demonstrate Computer Hardware and Install software.
Year:2022-2023(ODD) SEMESTER III		
MA1303A - Discrete Mathematics		
18	C201.1	K1 Describe the concepts needed to test the logic of a program.
	C201.2	K2 Explain the counting principles.
	C201.3	K2 Discuss Graph Theory that would help them to define new levels of networks which are implemented in AI and ANN.
	C201.4	K3 Demonstrate the properties of algebraic structures such as groups, rings and fields.
	C201.5	K3 Solve the Lattices and Boolean algebra used in Coding and Decoding theory of Cryptography.
IT1301A - Object Oriented Programming		
19	C202.1	K3 Demonstrate Java programs using OOP principles.
	C202.2	K3 Illustrate Java programs using inheritance, interfaces.
	C202.3	K3 Show Java applications using Exceptions and I/O streams.
	C202.4	K2 Describe the Java applications with Threads, generics classes and JDBC
	C202.5	K3 Apply interactive Desktop application using Swing and JDBC.
CS1301A - Data Structures		
20	C203.1	K3 Apply the concept and operations of List ADT.
	C203.2	K2 Explain the concept and operations of Stack and Queue ADT
	C203.3	K3 Demonstrate the various Tree ADT and its Applications
	C203.4	K3 Apply Graph data structures and hashing concept in real-world scenarios
	C203.5	K2 Interpret the implementation of sorting and searching
EC1306A - Digital Principles and System Design		
21	C204.1	K2 Discuss Boolean functions using K-Map
	C204.2	K2 Describe the Combinational Circuits
	C204.3	K2 Explain the Sequential Circuits.
	C204.4	K2 Summarize the designs using Programmable Logic Devices
	C204.5	K3 Apply HDL code for combinational and Sequential Circuits
CS1302A - Software Engineering		
	C205.1	K2 Describe the projects through the adoption of suitable lifecycle models
	C205.2	K3 Determine an appropriate process model depending on the user requirements
	C205.3	K3 Examine the suitable software design with appropriate user and component level design.
	C205.4	K3 Apply a suitable testing strategy for the designed software product.

22	C205.5	K3	Illustrate appropriate project estimation techniques and manage the risk.
IT1302A - Object Oriented Programming Laboratory			
	C206.1	K3	Illustrate and implement Java programs for simple applications that make use of classes, packages, inheritance and interfaces.
	C206.2	K3	Demonstrate and implement Java programs with array list and exception handling
	C206.3	K3	Illustrate the concept of Multithreading using java programs.
	C206.4	K3	Examine the applications using file processing, generic programming.
23	C206.5	K3	Demonstrate the applications using event handling mechanism.
CS1303A - Data Structures Laboratory			
	C207.1	K3	Demonstrate appropriate operations and applications of Linear datastructures(List, Stack and Queue) using array
	C207.2	K3	Examine functions to implement operations of Linear datastructures(List, Stack and Queue) using Linked List
	C207.3	K3	Determine the different operations of various Trees
	C207.4	K3	Demonstrate the application of graph traversal algorithms and techniques
24	C207.5	K3	Examine various sorting, searching and hashing algorithms
HS1301A - Interpersonal Skills Laboratory			
	C208.1	K2	Explain the communication skills in articulating a complete idea with a clarity in pronunciation
	C208.2	K2	Describe the interpersonal skills with a willingness to work in team following the social work ethics
	C208.3	K2	Discuss the speaking ability to respond in any informal talk detailing the content of the information in any form
	C208.4	K2	Explain the role of active listening and reading skills in group discussions to effectively persuade the group to achieve the desired
25	C208.5	K2	Explain the effective formal presentations and interactive communication in any academic and business contexts
Year:2022-2023(EVEN) SEMESTER IV			
MA1403A - Probability and Statistics			
	C209.1	K2	Describe the fundamental knowledge of the concepts of probability and have knowledge of standard distributions which can describe real life phenomenon.
	C209.2	K2	Explain the basic concepts of one and two dimensional random variables and apply in engineering applications.
	C209.3	K3	Apply the concept of testing of hypothesis for small and large samples in real life problems.
	C209.4	K3	Solve the basic concepts of classifications of design of experiments in the field of agriculture and statistical quality control.
26	C209.5	K3	Calculate the notion of sampling distributions and statistical techniques used in engineering
CS1401A - Computer Architecture			
	C210.1	K3	Examine the basic structure of computers, operations and instructions.
	C210.2	K4	Analyze the design of the arithmetic and logic unit.
	C210.3	K3	Demonstrate the pipelined execution and design of control unit
	C210.4	K3	Illustrate the parallel processing architectures
27	C210.5	K4	Analyze the various memory systems and I/O communication.
IT1401A - Database Management Systems			
	C211.1	K3	Illustrate data requirements using conceptual modeling and design database schemas based on the conceptual model and Formulate solutions to a broad range of query.

28	C211.2	K4	Analyze the components of normalization and evaluate how different normal forms impact the structure and efficiency of a database
	C211.3	K3	Demonstrate the transactions and estimate the procedures for controlling the consequences of concurrent data access
	C211.4	K4	Analyze the performance of different file storage models and indexing strategies, assessing their influence on the efficiency of query processing and optimization processes
	C211.5	K4	Illustrate the features of distributed, semi-structured, and unstructured database systems to evaluate how they address specific data storage and retrieval challenges.
CS1402A - Design and Analysis of Algorithms			
29	C212.1	K3	Apply efficient algorithms for both recursive and non-recursive relations to compute space and time complexity
	C212.2	K4	Analyze the different algorithms using design techniques like Brute force and Divide and Conquer
	C212.3	K3	Demonstrate algorithms using design paradigms like Greedy and Dynamic Programming for a given problem.
	C212.4	K3	Examine algorithms using Backtracking and Branch & Bound design techniques.
	C212.5	K4	Categorize various algorithms and interpret solutions to evaluate NP Hard problems
HV1401A - Universal Human Values			
30	C213.1	K2	Summarize the essentials of human values and skills, self-exploration, happiness and prosperity.
	C213.2	K3	Examine harmony in human being.
	C213.3	K3	Illustrate the role of harmony in family, society and universal order.
	C213.4	K2	Describe the holistic perception of harmony across all levels of existence.
	C213.5	K2	Explain the appropriate technologies and management patterns that help create harmony between professional and personal life.
CS1403A - Object Oriented Analysis and Design			
31	C214.1	K3	Apply the software design with UML diagrams
	C214.2	K3	Demonstrate the software applications using Object Oriented concepts
	C214.3	K3	Determine the various scenarios based on software requirements
	C214.4	K3	Experiment with UML-based software design by applying design patterns to develop pattern-based designs.
	C214.5	K3	Illustrate the various testing methodologies for Object Oriented software
IT1402A - Database Management Systems Laboratory			
32	C215.1	K3	Demonstrate the use of typical data definitions and manipulation commands.
	C215.2	K3	Illustrate the applications to test Nested and Join Queries
	C215.3	K3	Demonstrate simple applications that use Views, Sequences, Synonyms, Cursors and critically analyze the use of Tables, Functions, Procedures, Triggers and Exception Handling using PL/SQL
	C215.4	K3	Modify and manipulate the data using NOSQL database.
	C215.5	K3	Model database using ER modeling, Normalize and Implement a GUI application that require a Front-end and Back end Tool
CS1404A - Design and Analysis of Algorithms Laboratory			
	C216.1	K2	Interpret algorithms for given problem and trace its complexity.
	C216.2	K2	Summarize the importance of different algorithm for solving same problem
	C216.3	K4	Analyze efficient algorithms to get optimal solution for the problem

33	C216.4	K3	Apply the suitable algorithm for the given problem using backtracking method.
	C216.5	K2	Describe the different techniques used in approximation algorithms and their applications.
HS1401A - Employability And Soft Skills Laboratory			
34	C217.1	K2	Summarize the articulating ideas clearly in both general and technical communication skills in terms of clarity and pronunciation.
	C217.2	K2	Discuss best practices for crafting and presenting formal job applications.
	C217.3	K3	Describe Excel in Verbal aptitude, read and evaluate texts logically to solve the puzzles.
	C217.4	K3	Demonstrate the ability to prepare and present formal job applications effectively.
	C217.5	K3	Illustrate the use of analytical reasoning in addressing professional problems.
Year:2023-2024(ODD) SEMESTER V			
MA1501A - Algebra and Number Theory			
35	C301.1	K2	Describe the basic notions of groups, rings, fields which will then be used to solve related problems
	C301.2	K2	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts.
	C301.3	K2	Discuss accurate and efficient use of advanced algebraic techniques.
	C301.4	K2	Discuss their mastery by solving non – trivial problems related to the concepts, and by proving simple theorems about the, statements proven by the text.
	C301.5	K3	Apply integrated approach to number theory and abstract algebra, and provide a firm basis for further reading and study in the
IT1501A - Computer Networks			
36	C302.1	K2	Describe the basics of the network and discuss the functions of the physical layer.
	C302.2	K2	Summarize the basics of data flow from one node to another.
	C302.3	K3	Explain the various services and protocols of the network layer.
	C302.4	K3	Determine the services of different transport layer protocols.
	C302.5	K3	Examine the working of various application layer protocols.
CS1501A - Operating Systems			
37	C303.1	K2	Summarize the basic concepts, functions, processes and threads of operating systems.
	C303.2	K3	Analyze process scheduling, synchronization, and deadlock handling mechanisms to design efficient process management strategies
	C303.3	K3	Examine various memory management schemes.
	C303.4	K3	Illustrate the functionality of file systems.
	C303.5	K2	Interpret the administrative tasks on Linux Servers and compare iOS and Android Operating Systems.
CS1502A - Theory of Computation			
38	C304.1	K3	Show the construction of DFA,NFA and ϵ -NFA and Regular expression for the languages
	C304.2	K2	Interpret the concept of converting NFA to DFA, ϵ -NFA to DFA, automata to regular expressions.
	C304.3	K3	Demonstrate the pumping lemma for regular languages and context free languages
	C304.4	K3	Predict the construction of CFG,Parse trees and PDA.
	C304.5	K2	Summarize the working Principles of Turing Machine and Post correspondence problem
EC1515A - Embedded System			
	C305.1	K2	Describe the architecture and programming of ARM processor

39	C305.2	K2	Explain the concepts of embedded systems
	C305.3	K2	Describe the basic concepts of real time operating system design
	C305.4	K3	Use the system design techniques to develop software for embedded systems
	C305.5	K3	Demonstrate the general purpose operating system and the real time operating system
IT1503A - Networks Laboratory			
40	C306.1	K3	Illustrate the use of TCP and UDP protocols.
	C306.2	K3	Examine the performance of different transport layer protocols.
	C306.3	K3	Use simulation tools to determine the performance of various network protocols.
	C306.4	K3	Apply various routing algorithms.
	C306.5	K3	Demonstrate the use of simulation tools
CS1503A - Operating Systems Laboratory			
41	C307.1	K3	Illustrate the various UNIX commands, shell programming
	C307.2	K3	Demonstrate inter-process communication (IPC) and semaphores, and analyze the performance of various CPU scheduling
	C307.3	K3	Demonstrate Deadlock avoidance and Detection Algorithms
	C307.4	K3	Illustrate the performance of the various Page Replacement Algorithms
	C307.5	K3	Examine File Organization and File Allocation Strategies
Year:2023-2024(EVEN) SEMESTER VI			
42	CS1601A - Internet Programming		
	C308.1	K2	Summarize the basics of Web Designing using HTML, CSS, and XML
	C308.2	K3	Examine and implement dynamic web page with validation using Java Script and JSON
	C308.3	K4	Illustrate server side programs using Servlet and JSP.
	C308.4	K4	Infer simple web pages in PHP
43	C308.5	K4	Analyze interactive web applications using modern internet programming.
	CS1602A- Compiler Design		
	C309.1	K2	Demonstrate the ability to design a compiler given a set of language features
	C309.2	K3	Demonstrate the knowledge of patterns, tokens and regular expressions for lexical analysis.
	C309.3	K2	Acquire skills using lex tool & yacc tool for developing a scanner and parser
44	C309.4	K2	Design and implement LL and LR parsers and Design algorithms to decode optimization in order to improve the performance of a programming
	C309.5	K4	Design algorithms to generate machine code
	CS1603A - Artificial Intelligence		
	C310.1	K3	Illustrate the intelligent agent frameworks and its environments.
	C310.2	K4	Analyze real life problems and solve those using searching algorithms and AI techniques.
44	C310.3	K4	Infer and apply logical reasoning in solving AI problems using first order and predicate logic.
	C310.4	K3	Demonstrate the applications of Probabilistic Reasoning in Bayesian Networks.
	C310.5	K3	Examine the various applications of AI and types of Learning.

CS1604A - Cryptography and Network Security

Summarize the fundamentals of network security, security architecture, threats, and vulnerabilities
 Apply symmetric cryptographic algorithms to perform encryption and decryption
 Demonstrate public-key cryptographic operations for secure communication
 Illustrate and analyze authentication mechanisms to ensure secure access control in information systems.
 Examine the advanced encryption methods, including identity-based encryption and quantum cryptography techniques

45

C311.1 K2
 C311.2 K3
 C311.3 K3
 C311.4 K3
 C311.5 K3

CS1607A - Cyber Security

Describe the basic concepts of Cyber Security.
 Summarize the security applications for operating system and networks
 Interpret the security detection and classify their various security countermeasures.
 Infer the various applicable Privacy principles and policies
 Illustrate the security planning and effective management of cyber security

46

C312.1 K2
 C312.2 K2
 C312.3 K2
 C312.4 K4
 C312.5 K3

CS1608A - Software Testing

Summarize the fundamentals of software testing
 Describe and design test cases suitable for a software development for different domains
 Explain the various test cases and document test plans.
 Apply a wide variety of testing techniques in an effective and efficient manner
 Illustrate automatic testing tools in different domains

47

C313.1 K2
 C313.2 K3
 C313.3 K3
 C313.4 K3
 C313.5 K4

CS1605A - Internet Programming Laboratory

Demonstrate Web pages using HTML/XML and style sheets.
 Model Client side dynamic web pages with validation using Java Script and JSON
 Infer Server side dynamic web pages with validation using Servlet and JSP
 Illustrate simple dynamic and interactive Web pages using PHP.
 Analyze web applications using recent Tools

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C314.1 K3
 C314.2 K3
 C314.3 K4
 C314.4 K4
 C314.5 K4

CS1606A - Security Laboratory

Illustrate code for classical Encryption Techniques to solve the problems
 Infer cryptosystems by applying symmetric and public-key encryption algorithms.
 Analyze code for authentication algorithms.
 Model a signature scheme using the Digital Signature Standard.
 Demonstrate the network security system using open-source tools.

49

C315.1 K3
 C315.2 K4
 C315.3 K4
 C315.4 K3
 C315.5 K3

HS1601A - Professional Communication Laboratory

Summarize active listening with passive listening in communication scenarios.
 Interpret verbal and non-verbal cues to engage effectively in group discussion.
 Discuss strategies to plan, design, and deliver impactful presentations.
 Demonstrate confidence and clarity while interacting for both formal and informal contexts.
 Apply persuasive communication techniques in different conversational contexts.

50

C316.1 k2
 C316.2 k2
 C316.3 k2
 C316.4 k3
 C316.5 k3

Year:2024-2025(ODD) SEMESTER VII

IT1701A -Distributed Systems and Cloud Computing		
51	C401.1	K4
	C401.2	K4
	C401.3	K3
	C401.4	K4
	C401.5	K4
CS1701A-Machine Learning		
52	C402.1	K3
	C402.2	K4
	C402.3	K3
	C402.4	K3
	C402.5	K3
MG1701A-Principles of Management		
53	C403.1	K2
	C403.2	K2
	C403.3	K3
	C403.4	K4
	C403.5	K3
CS1707A-Human Computer Interaction		
54	C404.1	K2
	C404.2	K2
	C404.3	K3
	C404.4	K3
	C404.5	K3
CS1708A-C# and .Net Programming		
55	C405.1	K3
	C405.2	K3
	C405.3	K3
	C405.4	K3
	C405.5	K3
CS1711A -Full Stack Web Development		
56	C406.1	K2
	C406.2	K3
	C406.3	K3

56	C406.4	K3	Predict and manage NoSQL databases with MongoDB.
	C406.5	K4	Outline and deploy a full-stack Single Page Application (SPA) utilizing React, NodeJS, and MongoDB in a cloud environment.
CS1713A-Blockchain Technologies			
57	C407.1	K2	Summarize the basics of Blockchain
	C407.2	K3	Examine the differences between conventional distributed database and Blockchain
	C407.3	K3	Illustrate the Satoshi Nakamoto consensus and differences between proof-of-work and proof-of-stake consensus
	C407.4	K3	Show the various concepts used in Cryptocurrency and Ethereum
	C407.5	K3	Demonstrate the various Cryptocurrency regulation and applications.
CS1702A-Machine Learning Laboratory			
58	C408.1	K3	Illustrate supervised learning algorithms for real world dataset
	C408.2	K3	Apply the concept of unsupervised learning algorithms for suitable application.
	C408.3	K3	Use appropriate probabilistic methods for real time application
	C408.4	K3	Apply various tools Weka/MATLAB etc for implementing machine learning algorithms
	C408.5	K3	Demonstrate Machine Learning algorithms to solve real world problems
CS1703A-Summer Internship			
59	C409.1	K2	Summarize the basic concepts related to the assigned project work during industrial training/certification course.
	C409.2	K4	Analyze the assigned problem by considering its impact on society and environment
	C409.3	K4	Justify and discuss the design solution for the problem.
	C409.4	K3	Demonstrate the ability to work effectively in team with commitments to professional ethics.
	C409.5	K5	Conclude the project work through proper documentation in the form of project report.
IT1702A-Cloud Computing Laboratory			
60	C410.1	K3	Demonstrate the cloud, its characteristics, various delivery and deployment models
	C410.2	K3	Examine fundamental cloud components mechanism with which cloud data centers are managed and administered.
	C410.3	K4	Infer and deploy a web application in a PaaS & SaaS environment.
	C410.4	K4	Outline the strength of virtualization and its role in enabling the cloud computing model
	C410.5	K4	Illustrate Hadoop single node cluster and run applications with map reduce commands
CS1714A-Project Work I			
61	C411.1	K2	Interpret emerging technologies and apply to real industry-level use cases.
	C411.2	K2	Summarize agile development process.
	C411.3	K3	Demonstrate career readiness competencies, Team Skills/Leadership qualities.
	C411.4	K3	Model Time management. Project management skills and communication skills.
	C411.5	K3	Use critical thinking for innovative problem solving.
Year:2024-2025(EVEN) SEMESTER VIII			
GE1801A-Professional Ethics in Engineering			
	C412.1	K2	understand the human values with regard to the individual life style for the society
	C412.2	K3	Examine the role of ethics to the engineering field
	C412.3	K4	Categorize how engineering is applied in association with ethics based on engineering experimentation

62	C412.4	K3	Illustrate the engineering ethics-based safety, responsibilities and rights
	C412.5	K4	Classify the global issues of professional ethics in engineering.
CS1809A-UI & UX Design			
63	C413.1	K2	Summarize the Foundations of Design thinking
	C413.2	K3	Apply visual and UI Design Principles
	C413.3	K3	Illustrate the various UX Design Principles
	C413.4	K4	Outline wireframes, Prototypes, and Usability Tests
	C413.5	K3	Apply research, ideation, and information architecture principles to design effective and user-centered solutions.
CS1810A-Green Computing			
64	C414.1	K2	Interpret the knowledge to adopt green computing practices to minimize negative impacts on the environment.
	C414.2	K2	Explain the green assets and modeling.
	C414.3	K3	Illustrate the skill in energy saving practices in their use of hardware.
	C414.4	K3	Determine the technology tools that can reduce paper waste and carbon footprint by the stakeholders.
	C414.5	K3	Examine the ways to minimize equipment disposal requirements.
CS1813A-Digital Marketing			
65	C415.1	K2	Explain about the designing, developing and promoting through Blogs
	C415.2	K3	Apply the search engine techniques for better visibility in the online world
	C415.3	K3	Illustrate an e-mail / Mobile marketing for promotional activities
	C415.4	K3	Examine social media marketing techniques to build customer relationships.
	C415.5	K3	Determine digital transformation techniques in email, mobile and social media
CS1801A-Project Work-II			
66	C416.1	K3	Predict a real world challenging problem ethically.
	C416.2	K3	Illustrate the requirements of the project and analyse extensively the literature survey.
	C416.3	K3	Examine technical ideas, strategies and methodologies in written and oral presentations.
	C416.4	K5	Evaluate sustainable solutions by formulating proper methodology and implement them.
	C416.5	K4	Outline new tools, algorithms, and/or techniques that contribute to the software solution of the project

N. Q. L.
Process Owner

A. HoD

SAVA ENGINEERING COLLEGE, CHENNAI - 77
(An Autonomous Institution Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE OUTCOMES

2023-2024 BATCH

Sr.No	COURSE NAME	SKILL MAPPING	COURSE OUTCOMES
Year 2020-2021(OOD) SEMESTER I			
HS1101 - Technical English			
1	CS101.1	K4	Exhibit the memory of the previously learnt grammatical concepts and apply them in technical contexts
	CS101.2	K4	Indicate the professional proficiency by preparing the formal correspondence with global standards.
	CS101.3	K5	Inter information and correlate the technical and general texts
	CS101.4	K4	Relate verbal and technical information concisely with coherence and cohesion.
	CS101.5	K5	Identify and construct appropriate statements representing the global situations
MA1101 - Calculus and its applications			
2	CS102.1	K4	Acquire the basic concepts of differentiation rules to find the maxima and minima for function of one variable
	CS102.2	K3	Attain the knowledge of basic application problems involved in first order differential equation
	CS102.3	K4	Compute the solution of second order linear differential equations with constant coefficients.
	CS102.4	K5	Explore the concept of ordinary derivatives into partial derivatives and apply them to find extreme values of the functions of two variables and series approximation of the function of two variables.
	CS102.5	K4	Identify the double and Triple integration techniques over a region in two dimensional and three-dimensional geometry.
PH1101 - Applied Physics			
3	CS103.1	K2	The students will gain knowledge on the basics of properties of matter and its applications.
	CS103.2	K2	Use the concepts of waves and optical devices and their applications in Laser and fiber optics.
	CS103.3	K2	The students will understand the properties of thermal materials and its applications.
	CS103.4	K4	The students will get knowledge on advanced physics concepts of quantum theory and its application in one dimensional box.
	CS103.5	K2	The students will understand the different types of crystals structures and different crystal growth techniques.
CH1101 - Engineering Chemistry			
4	CS104.1	K4	Explain the hardness of water, related problems and its treatment.
	CS104.2	K3	Apply phase rule to construct phase diagram and predict the low melting alloys.
	CS104.3	K4	Illustrate conventional and non-conventional energy sources and basics of electrochemistry. Types of batteries, their reactions and the importance.
	CS104.4	K4	Explain various fuels and its applications based on its calorific value.

	C104.5	K4	Comprehend the basics, types, preparation methods and recent trends in nanomaterial.
	CS1101 - Problem Solving and Python Programming		
5	C105.1	K2	Understand the basic of algorithmic problem solving.
	C105.2	K3	Illustrate Python programs with data expressions and statements.
	C105.3	K3	Apply Python control flow and python functions.
	C105.4	K3	Apply Python data structures like list, tuples and dictionaries.
	C105.5	K3	Implement file concepts to read and write data from/to files.
	ME1101 - Engineering Graphics		
6	C106.1	K1	Understand existing national standards of engineering drawing and visualization concepts .
	C106.2	K1	Understand the projection of Points, Lines and Plane surfaces.
	C106.3	K1	Understand the projection of Solids.
	C106.4	K1	Understand the section of solids.
	C106.5	K1	Understand the development of surfaces.
	BS1101 - Physics and Chemistry Laboratory		
7	C107.1	K2	Understand and apply fundamental concepts of mechanical properties analysis.
	C107.2	K2	Understand the fundamental principles of acoustic properties, and learn about semiconductor properties.
	C107.3	K2	Understand and learn about the fundamental principles of optical properties analysis.
	C107.4	K3	Apply hands-on knowledge in the quantitative chemical analysis of water.
	C107.5	K3	Carry out the basics of instrumental analysis-conductivity meter, and potentiometer.
	CS1102 - Problem Solving and Python Programming Laboratory		
8	C108.1	K4	Design simple programs using conditionals and loops.
	C108.2	K3	Implement functions to solve mathematical problems.
	C108.3	K3	Use strings for structuring Python programs.
	C108.4	K2	Summarize compound data using Python lists, tuples, and dictionaries.
	C108.5	K3	Implement file concepts to write programs to read and write data.
	Year:2020-2021(EVEN) SEMESTER II		
	HS1201- English for Communication		
9	C109.1	K1	Sketch brief technical and general texts using appropriate diction.
	C109.2	K1	Express various styles of drafts in formal and informal contents.
	C109.3	K3	Inter relate the verbal and technical skills in the given technical contexts.
	C109.4	K3	Construct meaningful utterances to present intense technical knowledge.

	C109.5	K4	Interpret key ideas by reviewing lexicon across the language in syntactical contexts.
	MIA1201 - Complex Variables and Transforms		
10	C110.1	K5	Understanding the concept of divergence and curl to Calculate the line, surface and volume integrals using vector integration.
	C110.2	K4	Solve the problems under analytic functions and construction of analytic function using C-R equation.
	C110.3	K4	Classify the singularities and pole, find residues and evaluate complex integrals using residue theorem.
	C110.4	K4	Understand the concepts of Laplace Transforms and its properties and to solve the differential equations.
	C110.5	K4	Using the properties of Z-transform and solve the difference equation.
	PH11201 - Materials Science		
11	C111.1	K4	Understand the fundamental principles of conducting materials and analyze their behavior under varying temperature conditions.
	C111.2	K4	Gain knowledge on semiconductor physics, including carrier concentration, and apply it to the design and analysis of semiconductor devices.
	C111.3	K4	Understand the properties of magnetic and superconducting materials, and apply their concepts in practical engineering systems.
	C111.4	K4	Gain knowledge on polarization mechanisms in dielectric materials and its applications.
	C111.5	K5	Understand the properties and applications of advanced nano materials and biomaterials for various engineering applications.
	EE1202- Basic Electrical, Electronics and Measurement Engineering		
12	C112.1	K1	Understand and analyse the fundamental laws & electrical network theorems.
	C112.2	K1	Gain knowledge on DC & AC static and rotating machines.
	C112.3	K1	Gain knowledge on renewable energy sources, various electrical protective devices.
	C112.4	K2	To understand the fundamentals of electronic circuits.
	C112.5	K2	To gain knowledge on the working principle & operation of measuring instruments and transducers.
	CS1201 - Programming in C		
13	C113.1	K2	Understand the basic syntax and semantics of the programming language constructs.
	C113.2	K3	Develop derived data types like arrays in solving problem.
	C113.3	K3	Solve a problem into modules and reconstruct it using various ways of user-defined functions.
	C113.4	K3	Develop user-defined data types like structures and unions and its applications to solve problems.
	C113.5	K4	Design applications using sequential and random-access file processing.
	CS1202 - Fundamentals of Computing		
14	C114.1	K2	Understand the major components of a computer system and architecture.
	C114.2	K4	Compare and contrast various indexing strategies in different database systems.
	C114.3	K2	Understand the programming languages and Operating System concepts.
	C114.4	K4	Design websites that meet specified needs and interests using basic elements to control layout.

	CL14.5	K.2	Understand the different uses of the infinitive
	Section 14.5: The infinitive		
14	CL14.5.1	K.3	Describe the different uses of the infinitive in English
	CL14.5.2	K.3	Describe the different uses of the infinitive in English
	CL14.5.3	K.3	Describe the different uses of the infinitive in English
	CL14.5.4	K.3	Describe the different uses of the infinitive in English
	CL14.5.5	K.3	Describe the different uses of the infinitive in English
	Section 14.6: The infinitive		
14	CL14.6.1	K.3	Understand the basic principles of writing
	CL14.6.2	K.3	Understand the basic principles of writing
	CL14.6.3	K.3	Understand the basic principles of writing
	CL14.6.4	K.3	Understand the basic principles of writing
	CL14.6.5	K.3	Understand the basic principles of writing
	Section 14.7: The infinitive		
14	CL14.7.1	K.3	Understand the basic principles of writing
	CL14.7.2	K.3	Understand the basic principles of writing
	CL14.7.3	K.3	Understand the basic principles of writing
	CL14.7.4	K.3	Understand the basic principles of writing
	CL14.7.5	K.3	Understand the basic principles of writing
	Section 14.8: The infinitive		
14	CL14.8.1	K.3	Understand the basic principles of writing
	CL14.8.2	K.3	Understand the basic principles of writing
	CL14.8.3	K.3	Understand the basic principles of writing
	CL14.8.4	K.3	Understand the basic principles of writing
	CL14.8.5	K.3	Understand the basic principles of writing
	Section 14.9: The infinitive		
14	CL14.9.1	K.3	Understand the basic principles of writing
	CL14.9.2	K.3	Understand the basic principles of writing
	CL14.9.3	K.3	Understand the basic principles of writing
	CL14.9.4	K.3	Understand the basic principles of writing
	CL14.9.5	K.3	Understand the basic principles of writing
	Section 14.10: The infinitive		
14	CL14.10.1	K.3	Understand the basic principles of writing
	CL14.10.2	K.3	Understand the basic principles of writing
	CL14.10.3	K.3	Understand the basic principles of writing
	CL14.10.4	K.3	Understand the basic principles of writing
	CL14.10.5	K.3	Understand the basic principles of writing

	C202.5	K3	Develop interactive Desktop application using Swing and JDBC.
CS1301 - Data Structures			
20	C203.1	K3	Apply the concept and operations of List ADT.
	C203.2	K2	Understand and apply the concept and operations of Stack and Queue ADT.
	C203.3	K3	Demonstrate the various Tree ADT and its Applications.
	C203.4	K3	Apply Graph data structures and hashing concept in real-world scenarios.
	C203.5	K2	Interpret the implementation of sorting and searching.
EC1306 - Digital Principles and System Design			
21	C204.1	K1	Simplify Boolean functions using K-Map.
	C204.2	K2	Design and Analyze Combinational Circuits.
	C204.3	K2	Design and Analyze Sequential Circuits.
	C204.4	K2	Implement designs using Programmable Logic Devices.
	C204.5	K3	Write HDL code for combinational and Sequential Circuits.
CS1302 - Software Engineering			
22	C205.1	K2	Develop the project by adopting suitable lifecycle models.
	C205.2	K3	Determine an appropriate process model depending on the user requirements.
	C205.3	K3	Identify the suitable software design with appropriate user and component level design.
	C205.4	K3	Apply a suitable testing strategy for the designed software product.
	C205.5	K3	Apply appropriate project estimation techniques and manage the risk.
IT1302 -Object Oriented Programming Laboratory			
23	C206.1	K3	Develop and implement Java programs for simple applications that make use of classes, packages, inheritance and interfaces.
	C206.2	K3	Develop and implement Java programs with array list and exception handling.
	C206.3	K3	Develop and implement Java programs using the concept of Multithreading.
	C206.4	K3	Design and develop the applications using file processing, generic programming.
	C206.5	K3	Design and develop the applications using event handling mechanism.
CS1303 - Data Structures Laboratory			
24	C207.1	K3	Demonstrate appropriate operations and applications of Linear datastructures(List, Stack and Queue) using array.
	C207.2	K3	Examine functions to implement operations of Linear datastructures(List, Stack and Queue) using Linked List.
	C207.3	K3	Determine the different operations of various Trees.
	C207.4	K3	Apply graph traversal algorithms and techniques.
	C207.5	K3	Examine various sorting, searching and hashing algorithms.

ET11001 - Environmental Health Laboratory

23	C208.1	K.1	Describe the environmental health of a community or region with a focus on the environment.
	C208.2	K.1	Describe environmental health with a focus on the environment and the environment.
	C208.3	K.2	Describe the environmental health of a community or region with a focus on the environment and the environment.
	C208.4	K.2	Describe the environmental health of a community or region with a focus on the environment and the environment.
	C208.5	K.3	Describe the environmental health of a community or region with a focus on the environment and the environment.

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ET11001 - Environmental Health Laboratory

24	C209.1	K.1	Describe the environmental health of a community or region with a focus on the environment.
	C209.2	K.2	Describe the environmental health of a community or region with a focus on the environment.
	C209.3	K.3	Describe the environmental health of a community or region with a focus on the environment.
	C209.4	K.3	Describe the environmental health of a community or region with a focus on the environment.
	C209.5	K.3	Describe the environmental health of a community or region with a focus on the environment.

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27	C210.1	K.3	Describe the environmental health of a community or region with a focus on the environment.
	C210.2	K.4	Describe the environmental health of a community or region with a focus on the environment.
	C210.3	K.4	Describe the environmental health of a community or region with a focus on the environment.
	C210.4	K.5	Describe the environmental health of a community or region with a focus on the environment.
	C210.5	K.4	Describe the environmental health of a community or region with a focus on the environment.

ET11001 - Environmental Health Laboratory

29	C211.1	K.3	Describe the environmental health of a community or region with a focus on the environment.
	C211.2	K.4	Describe the environmental health of a community or region with a focus on the environment.
	C211.3	K.4	Describe the environmental health of a community or region with a focus on the environment.
	C211.4	K.4	Describe the environmental health of a community or region with a focus on the environment.
	C211.5	K.4	Describe the environmental health of a community or region with a focus on the environment.

ET11001 - Environmental Health Laboratory

30	C212.1	K.3	Describe the environmental health of a community or region with a focus on the environment.
	C212.2	K.3	Describe the environmental health of a community or region with a focus on the environment.

29	C212.3	K3	Demonstrate algorithms using design paradigms like Greedy and Dynamic Programming for a given problem.
	C212.4	K3	Examine algorithms using Backtracking and Branch & Bound design techniques.
	C212.5	K4	Analyze various algorithms and interpret solutions to evaluate NP Hard problems.
HV1401 - Universal Human Values			
30	C213.1	K2	Understand the essentials of human values and skills, self-exploration, happiness and prosperity.
	C213.2	K3	Examine harmony in human being.
	C213.3	K3	Illustrate the role of harmony in family, society and universal order.
	C213.4	K4	Categorize the holistic perception of harmony at all levels of existence.
	C213.5	K4	Infer appropriate technologies and management patterns to create harmony in professional and personal lives.
CS1403 - Object Oriented Analysis and Design			
31	C214.1	K3	Apply the software design with UML diagrams.
	C214.2	K3	Demonstrate the software applications using Object Oriented concepts.
	C214.3	K3	Determine the various scenarios based on software requirements.
	C214.4	K3	Experiment the UML based software design into pattern based design using design patterns.
	C214.5	K3	Illustrate the various testing methodologies for Object Oriented software.
IT1402 - Database Management Systems Laboratory			
32	C215.1	K3	Use typical data definitions and manipulation commands.
	C215.2	K3	Design applications to test Nested and Join Queries.
	C215.3	K3	Implement simple applications that use Views, Sequences, Synonyms, Cursors and critically analyze the use of Tables, Functions, Procedures, Triggers and Exception Handling using PL/SQL.
	C215.4	K3	Create and manipulate data using NOSQL database.
	C215.5	K3	Design database using ER modeling, Normalize and Implement a GUI application that require a Front-end and Back end Tool.
CS1404 - Design and Analysis of Algorithms Laboratory			
33	C216.1	K2	Interpret algorithms for given problem and trace its complexity.
	C216.2	K2	Understand the importance of different algorithm for solving same problem.
	C216.3	K4	Analyze efficient algorithms to get optimal solution for the problem.
	C216.4	K3	Apply the suitable algorithm for the given problem using backtracking method.
	C216.5	K2	Understand the approximation algorithms.
HS1401 - Employability And Soft Skills Laboratory			
	C217.1	K2	Write for different purposes in general and technical context skills in articulating a complete idea with a clarity in pronunciation.
	C217.2	K2	Write formal job applications.

34	C217.3	K4	Excel in Verbal aptitude, read and evaluate texts logically to solve the puzzles.
	C217.4	K3	Develop and demonstrate the employability and soft skills.
	C217.5	K2	Display critical thinking in various professional contexts.
Year:2022-2023(ODD) SEMESTER V			
MA1501 - Algebra and Number Theory			
35	C301.1	K1	Apply the basic notions of groups, rings, fields which will then be used to solve related problems.
	C301.2	K2	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts.
	C301.3	K2	Demonstrate accurate and efficient use of advanced algebraic techniques.
	C301.4	K2	Demonstrate their mastery by solving non – trivial problems related to the concepts, and by proving simple theorems about the, statements proven by the same.
	C301.5	K3	Apply integrated approach to number theory and abstract algebra, and provide a firm basis for further reading and study in the subject.
IT1501 - Computer Networks			
36	C302.1	K2	Describe the basics of the network and discuss the functions of the physical layer.
	C302.2	K2	Summarize the basics of data flow from one node to another.
	C302.3	K3	Demonstrate the various services and protocols of the network layer.
	C302.4	K3	Determine the services of different transport layer protocols.
	C302.5	K3	Examine the working of various application layer protocols.
CS1501 - Operating Systems			
37	C303.1	K2	Understand the basic concepts, functions, processes and threads of operating systems.
	C303.2	K3	Analyze various scheduling algorithms, Understand deadlock, prevention and avoidance algorithms.
	C303.3	K3	Compare and contrast various memory management schemes.
	C303.4	K3	Illustrate the functionality of file systems.
	C303.5	K2	Perform administrative tasks on Linux Servers and compare iOS and Android Operating Systems.
CS1502 - Theory of Computation			
38	C304.1	K3	Understand DFA, NFA, and their equivalence with epsilon transitions.
	C304.2	K2	Understand regular expressions and minimize finite automata.
	C304.3	K1	Analyze CFGs, parse trees, and apply normal forms.
	C304.4	K3	Design PDAs and Turing Machines and relate them to grammars.
	C304.5	K2	Apply the concept of undecidability to identify problems
EC1515 - Embedded System			
	C305.1	K2	Describe the architecture and programming of ARM processor.
	C305.2	K4	Outline the concepts of embedded systems.

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44	C310.3	K4	Analyze and apply logical reasoning in solving AI problems using first order and predicate logic.
	C310.4	K3	Demonstrate the applications of Probabilistic Reasoning in Bayesian Networks.
	C310.5	K3	Examine the various applications of AI and types of Learning.
CS1604 - Cryptography and Network Security			
45	C311.1	K2	Understand the fundamentals of network security, security architecture, threats, and vulnerabilities.
	C311.2	K3	Apply symmetric cryptographic algorithms to perform encryption and decryption.
	C311.3	K3	Demonstrate public-key cryptographic operations for secure communication.
	C311.4	K3	Implement and analyze authentication mechanisms to ensure secure access control in information systems.
	C311.5	K3	Experiment with advanced encryption methods, including identity-based encryption and quantum cryptography techniques.
CS1607 - Cyber Security			
46	C312.1	K1	Acquire the basic concepts of Cyber Security.
	C312.2	K2	Determine the security applications for operating system and networks.
	C312.3	K2	Discover the security detection and classify their various security counter measures.
	C312.4	K2	Infer the various applicable Privacy principles and policies.
	C312.5	K2	Identify security planning and effective management of cyber security.
CS1608 - Software Testing			
47	C313.1	K2	Understand the fundamentals of software testing.
	C313.2	K3	Design test cases suitable for a software development for different domains.
	C313.3	K3	Design test cases and document test plan.
	C313.4	K4	Apply a wide variety of testing techniques in an effective and efficient manner.
	C313.5	K4	Apply automatic testing tools in different domains.
CS1605 - Internet Programming Laboratory			
48	C314.1	K4	Design and Implement Web pages using HTML/XML and style sheets.
	C314.2	K4	Develop Client side dynamic web pages with validation using Java Script and JSON.
	C314.3	K4	Develop Server side dynamic web pages with validation using Servlet and JSP.
	C314.4	K4	Construct simple dynamic and interactive Web pages using PHP.
	C314.5	K4	Develop interactive web applications using recent Tools.
CS1606 - Security Laboratory			
49	C315.1	K3	Develop code for classical Encryption Techniques to solve the problems.
	C315.2	K4	Build cryptosystems by applying symmetric and public-key encryption algorithms.
	C315.3	K4	Construct code for authentication algorithms.

	C315.4	K4	Develop a signature scheme using the Digital Signature Standard.
	C315.5	K3	Demonstrate the network security system using open-source tools.
	HS1601 – Professional Communication Laboratory		
	C316.1	K1	Listen and respond appropriately.
	C316.2	K1	Participate in group discussions.
	C316.3	K2	Make effective presentations.
50	C316.4	K3	Participate confidently and appropriately in conversations both formal and informal.
	C316.5	K3	Persuade through conversations.
	Year:2023-2024(ODD) SEMESTER VII		
	IT1701 -Distributed Systems and Cloud Computing		
	C401.1	K2	Describe the concepts of distributed systems and Remote procedure calls.
	C401.2	K2	Able to understand real time distributed systems and load balancing approaches.
51	C401.3	K3	Elucidate the concepts of NIST Cloud Computing architecture and its design challenges.
	C401.4	K3	Implement different types of Virtualization technologies.
	C401.5	K4	Analyse the issues in Resource provisioning and Security governance in clouds.
	CS1701-Machine Learning		
	C402.1	K2	Understand the basic concepts of machine learning and its types.
	C402.2	K4	Solve the problems of different applications using supervised learning algorithms.
52	C402.3	K4	Analyze typical Unsupervised learning algorithms for different types of applications.
	C402.4	K3	Demonstrate the various Reinforcement algorithms.
	C402.5	K3	Examine probabilistic methods for real world applications.
	MG1701-Principles of Management		
	C403.1	K2	Understand about the planning and decision making process of management.
	C403.2	K2	Understand the organization structure, HR planning and control.
53	C403.3	K3	Apply functions of management like motivation, leadership and communication.
	C403.4	K4	Analyze the performance controlling process, techniques of control and reporting to the management.
	C403.5	K3	Apply principles of management in order to execute the role as a manager in IT industry.
	CS1707-Human Computer Interaction		
	C404.1	K2	Learn the effective dialog for HCI.
	C404.2	K3	Design effective HCI for individuals and persons with disabilities.
54	C404.3	K2	Understand the importance of user feedback.

	C404.4	K2	Comprehend the HCI implications for designing multimedia/ e-commerce/ e-learning Websites.
	C404.5	K3	Develop meaningful user interface.
CS1708-C# and .Net Programming			
55	C405.1	K3	Develop a simple applications using C# and describe the .Net framework.
	C405.2	K3	Develop various applications using C# advanced features in .Net Framework.
	C405.3	K3	Develop distributed applications by implementing base class libraries and database connectivity using ADO.Net.
	C405.4	K3	Develop window based applications using .Net Framework.
	C405.5	K3	Design a mobile application using .NET compact framework.
CS1711 - Full Stack Web Development			
56	C406.1	K2	Utilize CSS and JavaScript for effective client-side scripting.
	C406.2	K3	Design and implement the client-side architecture of web applications.
	C406.3	K3	Develop web applications using Node JS.
	C406.4	K3	Design and manage NoSQL databases with Mongo DB.
	C406.5	K4	Build and deploy a full-stack Single Page Application (SPA) utilizing React, Node JS, and Mongo DB in a cloud environment.
CS1713-Blockchain Technologies			
57	C407.1	K2	Understand the basics of Blockchain.
	C407.2	K3	Examine the differences between conventional distributed database and Blockchain.
	C407.3	K3	Illustrate the Satoshi Nakamoto consensus and differences between proof-of-work and proof-of-stake consensus.
	C407.4	K3	Show the various concepts used in Cryptocurrency and Ethereum.
	C407.5	K3	Demonstrate the various Cryptocurrency regulation and applications.
CS1702-Machine Learning Laboratory			
58	C408.1	K3	Implement supervised learning algorithms for real world dataset.
	C408.2	K3	Apply the concept of unsupervised learning algorithms for suitable application.
	C408.3	K3	Make use of appropriate probabilistic methods for real time application.
	C408.4	K3	Apply various tools Weka/MATLAB etc for implementing machine learning algorithms.
	C408.5	K3	Implement Machine Learning algorithms to solve real world problems.
CS1703-Summer Internship			
59	C409.1	K2	Understand the basic concepts related to the assigned project work during industrial training/certification course.
	C409.2	K4	Analyze the assigned problem by considering its impact on society and environment.
	C409.3	K5	Justify and discuss the design solution for the problem.
	C409.4	K3	Demonstrate the ability to work effectively in team with commitments to professional ethics.

	C409.5	K5	Conclude the project work through proper documentation in the form of project report.
	IT1702-Cloud Computing Laboratory		
60	C410.1	K3	Demonstrate the cloud, its characteristics, various delivery and deployment models.
	C410.2	K4	Relate fundamental cloud components mechanism with which cloud data centers are managed and administered.
	C410.3	K6	Design and deploy a web application in a PaaS & SaaS environment.
	C410.4	K6	Outline the strength of virtualization and its role in enabling the cloud computing model.
	C410.5	K5	Validate Hadoop single node cluster and run applications with map reduce commands.
	ECS1601 - Professional Readiness for Innovation, Employability and Entrepreneurship		
61	C411.1	K2	Upskill in emerging technologies and apply to real industry-level use cases.
	C411.2	K2	Understand agile development process.
	C411.3	K3	Develop career readiness competencies, Team Skills Leadership qualities.
	C411.4	K3	Develop Time management, Project management skills and communication skills.
	C411.5	K5	Use critical thinking for innovative problem solving.
	Year:2023-2024(EVEN) SEMESTER VIII		
	GE1801-Professional Ethics in Engineering		
62	C412.1	K2	Understand the human values with regard to the individual life style for the society.
	C412.2	K3	Examine the role of ethics to the engineering field.
	C412.3	K4	Categorize how engineering is applied in association with ethics based on engineering experimentation.
	C412.4	K3	Illustrate the engineering ethics-based safety, responsibilities and rights.
	C412.5	K4	Classify the global issues of professional ethics in engineering.
	CS1808-UI and UX Design		
63	C413.1	K2	Understand the Foundations of Design thinking.
	C413.2	K3	Apply Visual and UI Design Principles.
	C413.3	K3	Illustrate and Define UX Design Principles.
	C413.4	K4	Outline Wireframes, Prototypes, and Usability Tests.
	C413.5	K3	Apply Research, Ideation, and Information Architecture.
	CS1809 -Green Computing		
64	C414.1	K1	Acquire knowledge to adopt green computing practices to minimize negative impacts on the environment.
	C414.2	K2	Understand the green assets and modeling.
	C414.3	K2	Enhance the skill in energy saving practices in their use of hardware.
	C414.4	K2	Evaluate technology tools that can reduce paper waste and carbon footprint by the stakeholders.

	C414.5	K3	Understand the ways to minimize equipment disposal requirements.
	CS1813-Digital Marketing		
	C415.1	K2	Understand about the designing, developing and promoting through Blogs
	C415.2	K3	Apply the search engine techniques for better visibility in the online world
	C415.3	K3	Develop an e-mail / Mobile marketing for promotional activities
	C415.4	K3	Apply social media marketing techniques to build customer relationships.
65	C415.5	K3	Apply digital transformation techniques in email, mobile and social media
	CS1801-Project Work		
	C416.1	K3	Formulate a real world challenging problem ethically.
	C416.2	K3	Develop the requirements of the project and analyse extensively the literature survey.
	C416.3	K3	Express technical ideas, strategies and methodologies in written and oral presentations.
	C416.4	K5	Develop sustainable solutions by formulating proper methodology and implement them.
66	C416.5	K4	Self-learn new tools, algorithms, and/or techniques that contribute to the software solution of the project


 Process Owner


 HOD

S.A ENGINEERING COLLEGE, CHENNAI-77
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COURSE OUTCOMES

2019-2023 BATCH		COURSE OUTCOMES	
SL.No	NAME	MAPPING	
Year:2019-2020(ODD) SEMESTER I			
HS8151 - Communicative English			
1	C101.1	K1	Participate effectively in informal conversation: introduce themselves and their friends and express opinions in English
	C101.2	K1	Comprehend conversations and short talks delivered in English.
	C101.3	K2	Read and write in a coherent and organized manner
	C101.4	K2	Read articles of a general kind in and newspapers
	C101.5	K2	Write short essays of a general kind and personal letters and emails in English.
	C101.6	K2	Can communicate any idea of a general kind in an organized manner.
MA8151 - Engineering Mathematics - I			
2	C102.1	K1	Learn the basic concepts to use both the limit definition and rules of differentiation to differentiate functions.
	C102.2	K1	Apply partial differentiation to solve maxima and minima problems.
	C102.3	K2	techniques of integration, such as substitution, partial fractions and integration by parts
	C102.4	K1	Determine convergence/divergence of improper integrals and evaluate convergent improper integrals
	C102.5	K2	Apply integration to compute multiple integrals, area, volume, integrals in polar coordinates, In addition to change of order and change of variables
	C102.6	K1	Apply various techniques in solving differential equations.
PI18151 - Engineering Physics			
3	C103.1	K1	Gain knowledge on the basics of properties of matter and its applications
	C103.2	K1	Acquire knowledge on the concepts of waves and optical devices and their applications in fiber optics
	C103.3	K2	Get Adequate knowledge on the concepts of thermal properties of materials and their applications in expansion joints and heat exchangers
	C103.4	K1	Get knowledge on advanced physics concepts of quantum theory and its applications in tunneling microscopes
	C103.5	K2	Understand the basics of crystals, their structures and different crystal growth techniques
	C103.6	K1	Acquire knowledge on solid state laser, graphite and zinc blende structures.
CY8151 - Engineering Chemistry			
4	C104.1	K1	About the type of troubles faced in boilers and the methods used to treat impure water.
	C104.2	K1	The factors affecting the rate of adsorption and catalytic activity.
	C104.3	K1	With the help of phase rule, they could understand the various phase diagram and able to predict the low melting alloys
	C104.4	K1	Types of solid, liquid and gaseous fuel, manufacturing methods and basic steps involved in combustion reaction
	C104.5	K1	Types of batteries, their reactions and the importance of renewable energy resources.
	C104.6	K2	The basics and applications of nanomaterial in the recent innovations.
GE8151 - Problem Solving and Python Programming			
	C105.1	K1	Understand the basic of algorithmic problem solving
	C105.2	K1	Be familiar with data expressions and statements

5	C105.3	K2	understand control flow and functions problems
	C105.4	K1	comprehend list, tuples and dictionaries.
	C105.5	K1	Read and write data from/to files in Python Programs
	C105.6	K2	Understand Object oriented programming concepts
GE8152 - Engineering Graphics			
6	C106.1	K1	Understand existing national standards of engineering drawing and visualization concepts
	C106.2	K1	Understand the projection of Points, Lines and Plane surfaces
	C106.3	K1	Understand the projection of Solids
	C106.4	K1	Understand the section of solids
	C106.5	K1	Understand the development of surfaces
	C106.6	K1	Understand the isometric and perspective projections
GE8161 - Problem Solving and Python Programming Laboratory			
7	C107.1	K2	Write, test, and debug simple Python programs.
	C107.2	K3	Implement Python programs with conditionals and loops.
	C107.3	K3	Develop Python programs step-wise by defining functions and calling them.
	C107.4	K2	Use Python lists, tuples, dictionaries for representing compound data.
	C107.5	K3	Read and write data from/to files in Python
	C107.6	K2	Design and implement a program to solve a real-world problem using the language idioms, data structure and standard library
BS8161 - Physics and Chemistry Laboratory			
8	C108.1	K1	Apply principles of optics, sound and thermal properties for engineering.
	C108.2	K2	Apply principles of elasticity for engineering applications.
	C108.3	K3	Analyse the viscosity of a liquid
	C108.4	K2	Apply hands-on knowledge in the quantitative chemical analysis of water.
	C108.5	K1	Carryout the basics of instrumental analysis-conductivity meter, potentiometer, pH meter.
	C108.6	K2	Analyse the total dissolved solids in water
Year:2019-2020(EVEN) SEMESTER II			
HS8251- Technical English			
9	C109.1	K1	Learning the concepts of applying differential equation in signals
	C109.2	K2	Listen and comprehend lectures and talks in their area of specialization successfully.
	C109.3	K2	Speak appropriately and effectively in varied formal and informal contexts.
	C109.4	K2	Write reports and writing job applications
	C109.5	K2	Write minutes of meeting and to take part in group discussion effectively.
	C109.6	K2	Attain fluency in English language to speak convincingly, express their opinions clearly, initiate a discussion, negotiate and comprehend using appropriate communicative strategies
MA8251 - Engineering Mathematics - II			
	C110.1	K1	Eigen values and Eigen vectors, diagonalization of a matrix, Symmetric matrices, Positive definite matrices and similar matrices
	C110.2	K2	Gradient, Divergence and curl of a vector; scalar point function and related identities, evaluation of line, surface and volume integrals using Gauss, Stokes' and Green's theorems and their verification.

10	C110.3	K2	Analytic functions, conformal mapping
	C110.4	K3	Complex Integration
	C110.5	K3	Laplace transform and inverse transform of simple functions, properties, various related theorems and application to differential equations with constant coefficients
	C110.6	K4	Learning the concepts of applying differential equation in signals
PH8252 - Physics for Information Science			
11	C111.1	K1	To gain knowledge on industrial application of DC motors
	C111.2	K1	Acquire knowledge on basics of semiconductor physics and its applications in various devices
	C111.3	K2	Get knowledge on magnetic properties of materials and their applications in data storage
	C111.4	K2	Have the necessary understanding on the functioning of optical materials for optoelectronics
	C111.5	K2	Understand the basics of quantum structures and their applications in carbon electronics.
	C111.6	K3	Gain knowledge on application of semiconducting materials and preparation of nano materials.
UE8255 - Basic Electrical, Electronics and Measurement Engineering			
12	C112.1	K1	Understand and analyse the fundamental laws & electrical network theorems
	C112.2	K1	Gain knowledge on DC & AC static and rotating machines
	C112.3	K1	Gain knowledge on renewable energy sources, various electrical protective devices
	C112.4	K2	To understand the fundamentals of electronic circuits
	C112.5	K2	To gain knowledge on the working principle & operation of measuring instruments and transducers.
	C112.6	K2	To gain knowledge on industrial application of DC motors
GE8291 - Environmental Science and Engineering			
13	C113.1	K1	Students will be able to understand the functions of ecosystems and appreciate the bio diversity.
	C113.2	K2	Students will be able to know the measures to control environmental pollution.
	C113.3	K1	Students will be able to understand the usage as well as the effects of over exploitation of natural resources.
	C113.4	K1	Students will have knowledge about finding technological, economic and political solutions to environmental problems with various Environmental Protection Act in mind.
	C113.5	K2	Students will be able to understand the interrelationship between population explosion and the environment and also role of IT in environment and human health.
	C113.6	K3	Students will be able to understand that Environmental problems can only be solved by Public participation in all aspects and cannot be solved by mere laws.
CS8251 - Programming in C			
14	C114.1	K2	Develop simple applications in C using basic constructs
	C114.2	K2	Design and implement applications using arrays and strings
	C114.3	K3	Develop and implement applications in C using functions and pointers
	C114.4	K3	Develop applications in C using structures.
	C114.5	K3	Design applications using sequential and random access file processing.
	C114.6	K4	Understand the concept of dynamic memory management and basic data structures such as arrays and linked list
GE8261 - Engineering Practices Laboratory			
15	C115.1	K1	Understand the basic principle of welding
	C115.2	K2	Understand the basic principle of Sheet Metal Work
	C115.3	K2	Understand the basic principle of joints

15	C115.4	K2	Understand the basic principle of Machine assembly practice
	C115.5	K2	Understand the basic principle of Drilling
	C115.6	K3	Understand the basic principle of turning
CS8261- C Programming Laboratory			
16	C116.1	K2	Read, understand and trace the execution of programs written in C language
	C116.2	K3	Develop conditional and iterative statements to write C programs
	C116.3	K3	Develop C programs for simple applications making use of basic constructs, arrays and strings
	C116.4	K3	Develop C programs involving functions, recursion, pointers, and structures
	C116.5	K2	Design applications using sequential and random access file processing.
16	C116.6	K4	Exercise user defined functions to solve real time problems
Year:2020-2021(ODD) SEMESTER III			
MA8351 - Discrete Mathematics			
17	C201.1	K1	Have knowledge of the concepts needed to test the logic of a program.
	C201.2	K2	Be aware of the counting principles
	C201.3	K2	Learn the concepts of Graph Theory that would help them to define new levels of networks which are implemented in AI and ANN.
	C201.4	K2	Be exposed to concepts and properties of algebraic structures such as groups, rings and fields
	C201.5	K3	Expose the concepts and properties of Lattices and Boolean algebra used in Coding and Decoding theory of Cryptography
17	C201.6	K4	Be proficient in Logical and Mathematical maturity and ability to deal with abstraction and to introduce most of the basic terminologies used in computer science courses and application of ideas to solve practical problems.
CS8351 - Digital Principles and System Design			
18	C202.1	K1	Simplify Boolean functions using K-Map
	C202.2	K2	Design and Analyze Combinational Circuits
	C202.3	K2	Design and Analyze Sequential Circuits
	C202.4	K2	Implement designs using Programmable Logic Devices
	C202.5	K3	Write HDL code for combinational and Sequential Circuits
	C202.6	K4	ASM chart for digital circuits
CS8391 - Data Structures			
19	C203.1	K2	Learn and apply the concept and operations of List ADT
	C203.2	K2	Understand and apply the concept and operations of Stack and Queue ADT
	C203.3	K2	Gain the knowledge about Tree ADT and its Applications
	C203.4	K3	Understand the concept, operations and applications of Graph ADT
	C203.5	K3	Understand and analyze sorting, searching and hashing Techniques
19	C203.6	K4	Comprehend Minimum Spanning Trees – and algorithms (Kruskal and Prim's algorithm)
CS8392 - Object Oriented Programming			
20	C204.1	K2	Understand the fundamentals of OOP using Java.
	C204.2	K2	Learn the concepts of inheritance, interfaces

20	C204.3	K2	Build Java applications using exceptions and I/O streams
	C204.4	K3	Develop Java applications with threads and generic classes
	C204.5	K3	Develop interactive Java programs using swing
	C204.6	K4	Learn the concepts of Applets and develop the interactive Java programs using Applets
EC 839A - Communication Engineering			
21	C205.1	K2	Learn the concepts of Applets and develop the interactive Java programs using Applets
	C205.2	K2	Apply analog and digital communication techniques
	C205.3	K2	Use data and pulse Communication techniques
	C205.4	K3	Analyze Noise and Error control coding
	C205.5	K3	Apply spread spectrum modulation techniques
	C205.6	K4	Analyze basic concepts of cellular communication
CS331 - Data Structures Laboratory			
22	C206.1	K2	Implement the operations of List, Stack and Queue using array
	C206.2	K2	Implement the operations of List, Stack and Queue using linked list
	C206.3	K2	Understand and implement the different operations of various Trees
	C206.4	K3	Implement graph traversal algorithms and techniques
	C206.5	K3	Understand and implement various sorting, searching and hashing algorithms
	C206.6	K4	Implement Minimum Spanning Trees algorithm
CS333 - Object Oriented Programming Laboratory			
23	C207.1	K2	Develop and implement Java programs for simple applications that make use of classes, packages and interfaces
	C207.2	K2	Develop and implement Java programs with array list and exception handling
	C207.3	K2	Develop and implement Java programs using the concept of Multithreading
	C207.4	K3	Design and develop the applications using file processing, generic programming
	C207.5	K3	Design and develop the applications using event handling mechanism
	C207.6	K4	Develop the mini-project using the concepts of Java Programming
CS332 - Digital Systems Laboratory			
24	C208.1	K1	Implement simplified combinational circuits using basic logic gates
	C208.2	K2	Implement combinational circuits using MSI devices
	C208.3	K2	Implement sequential circuits like registers and counters
	C208.4	K3	Simulate combinational circuits using HDL
	C208.5	K3	Simulate sequential circuits using HDL
	C208.6	K4	Implement various digital circuits
HS331 - Interpersonal Skills/Listening & Speaking			
25	C209.1	K1	Listen and respond appropriately
	C209.2	K2	Participate in group discussions
	C209.3	K2	Make effective presentations
	C209.4	K2	Participate confidently and appropriately in conversations both formal and informal

25	C209.5	K3	Persuade through conversations
	C209.6	K4	Improve pronunciation clearly.
Year:2020-2021(EVEN) SEMESTER IV			
MA8402 - Probability and Queuing Theory			
26	C210.1	K1	describe real life phenomenon
	C210.2	K2	Understand the basic concepts of one and two dimensional random variables and apply in engineering applications.
	C210.3	K3	Apply the concept of random processes in engineering disciplines
	C210.4	K3	Acquire skills in analyzing queuing models.
	C210.5	K2	Understand and characterize phenomenon which evolve with respect to time in a probabilistic manner
	C210.6	K3	science and engineering.
CS8491- Computer Architecture			
27	C211.1	K2	Understand the basics structure of computers, operations and instructions.
	C211.2	K2	Ability to Design arithmetic and logic unit.
	C211.3	K2	Understand pipelined execution and design control unit
	C211.4	K3	Understand parallel processing architectures
	C211.5	K4	Understand the various memory systems and I/O communication.
	C211.6	K3	Understand Homogeneous and heterogeneous processors
CS8492 - Database Management Systems			
28	C212.1	K2	Classify the modern and futuristic database applications based on size and complexity
	C212.2	K3	Map ER model to Relational model to perform database design effectively
	C212.3	K2	Write queries using normalization criteria and optimize queries
	C212.4	K3	Compare and contrast various indexing strategies in different database systems
	C212.5	K4	Appraise how advanced databases differ from traditional databases.
	C212.6	K4	Study about various open source databases.
CS8451 - Design and Analysis of Algorithms			
29	C213.1	K2	Analyze the time and space complexity of algorithms.
	C213.2	K2	Design algorithms for various computing problems (brute -force, divide and conquer)
	C213.3	K3	Analyze the different algorithm design techniques for a given problem.
	C213.4	K2	Understand the application of Iterative algorithm.
	C213.5	K3	Modify existing algorithms to improve efficiency.
	C213.6	K4	Adopt the recent algorithm design technique to solve the given problem.
CS8493 - Operating Systems			
	C214.1	K2	Prepare reports for real life database applications
	C214.2	K2	Understand the principles of concurrency, design various CPU scheduling algorithms and Deadlock mechanisms.
	C214.3	K3	Compare and contrast various memory management schemes.
	C214.4	K4	Understand the prototype for file systems
	C214.5	K3	Perform administrative tasks on Linux Servers and compare iOS and Android OS.

30	C214.6	K4	Acquire knowledge about Operating systems used for business applications
CS8494 - Software Engineering			
	C215.1	K2	Identify key activities in software process and compare different process models
	C215.2	K2	Understand the concepts of requirement engineering and analysis modeling
	C215.3	K4	Applying the systematic procedure for software design
	C215.4	K3	Compare and Contrast various testing methods and implementation techniques
	C215.5	K3	Management techniques used for project schedule estimation of project cost
31	C215.6	K4	Understand the evolution of object oriented models
CS8491 - Database Management Systems Laboratory			
	C216.1	K2	Use typical data definitions and manipulation commands
	C216.2	K2	Design applications to test Nested and Join Queries
	C216.3	K3	Implement simple applications that use Views, Sequences, Synonyms and Cursors
	C216.4	K3	Design database using ER modeling, Normalize and Implement applications that require a Front-end Tool
	C216.5	K3	Critically analyze the use of Tables, Functions, Procedures, Triggers and Exception Handling using PL/SQL
32	C216.6	K4	Prepare reports for real life database applications
CS8461 - Operating Systems Laboratory			
	C217.1	K2	Compare the performance of various CPU Scheduling Algorithms
	C217.2	K2	Implement Deadlock avoidance and Detection Algorithms
	C217.3	K2	Implement Semaphores
	C217.4	K3	Create processes and implement IPC
	C217.5	K3	Analyze the performance of the various Page Replacement Algorithms
33	C217.6	K4	Implement File Organization and File Allocation Strategies
HS8461 - Advanced Reading and Writing			
	C218.1	K1	Write different types of essays
	C218.2	K1	Write winning job applications
	C218.3	K2	Read and evaluate texts critically
	C218.4	K2	Display critical thinking in various professional contexts
	C218.5	K3	Communicate effectively through mails
34	C218.6	K2	Justify the purpose of higher studies
Year:2021-2022(ODD) SEMESTER V			
MA8551 - Algebra and Number Theory			
	C301.1	K1	Apply the basic notions of groups, rings, fields which will then be used to solve related problems
	C301.2	K2	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts.
	C301.3	K2	Demonstrate accurate and efficient use of advanced algebraic techniques
	C301.4	K2	Demonstrate their mastery by solving non-trivial problems related to the concepts, and by proving simple theorems about the statements proven by the text
	C301.5	K3	Apply integrated approach to number theory and abstract algebra, and provide a firm basis for further reading and study in the subject

35	C301.6	K3	An ability to deal with abstraction and to introduce most of the basic concepts in number theory used in computer science courses and application in cryptography and network security
CS8591 - Computer Networks			
36	C302.1	K2	Understand the basic layers and its functions in computer networks
	C302.2	K2	Understand the basics of how data flows from one node to another
	C302.3	K3	Learn the functions of network layer and the various routing protocols
	C302.4	K3	Understand the functions and protocols of Transport layer
	C302.5	K3	Understand the working of various application layer protocols.
	C302.6	K3	Gain ideas on security issues of networks.
EC8691 - Microprocessors and Microcontrollers			
37	C303.1	K2	Understand the Architecture of 8086 microprocessor and ALP
	C303.2	K2	Design the bus architecture and multiprocessor configuration of 8086 microprocessor
	C303.3	K3	Design the I/O and Memory Interfacing circuits using 8086 and relevant case studies
	C303.4	K2	Understand the Architecture of 8051 microcontroller and ALP
	C303.5	K4	Design the various interfacing circuits using 8051
	C303.6	K3	Comprehend the Architecture of modern microprocessor
CS8501 - Theory of Computation			
38	C304.1	K2	Construct automata for any pattern
	C304.2	K2	Construct regular expression for any pattern
	C304.3	K3	Write Context free grammar for any construct.
	C304.4	K4	Design Turing machines for any language and propose computation solutions.
	C304.5	K3	Derive whether a problem is decidable or not.
	C304.6	K3	Understand Context Sensitive Grammar and LBA
CS8592 - Object Oriented Analysis and Design			
39	C305.1	K2	Express software design with UML diagrams
	C305.2	K4	Design software applications using OO concepts.
	C305.3	K3	Identify various scenarios based on software requirements.
	C305.4	K3	Transform UML based software design into pattern based design using design patterns.
	C305.5	K2	Understand the various testing methodologies for OO software.
	C305.6	K2	Compare and contrast various testing techniques.
OCE552- Geographic Information System			
40	C306.1	K1	Understand the basic idea about the fundamentals of GIS.
	C306.2	K2	Understand the types spatial of data models.
	C306.3	K2	Gain knowledge about data input and topology
	C306.4	K2	Gain knowledge on data analysis tools and network analysis
	C306.5	K3	Understand data management functions and data output.
	C306.6	K2	Understand the GIS in the Cloud.

CS8651 - Microprocessors and Microcontrollers I Laboratory		
41	C307.1	K2 Develop ALP Programs for fixed and Floating Point and Arithmetic using 8086.
	C307.2	K3 Interface different I/Os with 8086 processor
	C307.3	K2 Generate waveforms using 8086 Microprocessors
	C307.4	K3 Develop and execute programs using 8051 Microcontroller
	C307.5	K3 Develop and execute programs using MASM
	C307.6	K2 Study and Develop ALP program using ARM processor
CS8652 - Object Oriented Analysis and Design Laboratory		
42	C308.1	K2 Perform OO analysis and design for a given problem specification.
	C308.2	K2 Identify and map basic software requirements in UML mapping
	C308.3	K3 Improve the software quality using design patterns and to explain the rationale behind applying specific design patterns
	C308.4	K3 Test the compliance of the software with the SRS
	C308.5	K3 Generate code from design
	C308.6	K4 Design and develop a case study on object oriented projects
CS86581 - Networks Laboratory		
43	C309.1	K2 Implement various protocols using TCP and UDP
	C309.2	K2 Compare the performance of different transport layer protocols
	C309.3	K3 Use simulation tools to analyze the performance of various network protocols.
	C309.4	K4 Analyze various routing algorithms
	C309.5	K4 Implement error correction codes.
	C309.6	K2 Study about the configuration of Cisco Network Operating system.
Year:2021-2022(EVEN) SEMESTER VI		
CS8651 - Internet Programming		
44	C310.1	K2 Understand the java programming.
	C310.2	K2 Able to design a basic website using HTML and Cascading Style Sheets
	C310.3	K3 Able to Design and implement dynamic web page with validation using JavaScript objects and by applying different event handling mechanisms.
	C310.4	K3 Able to Design rich client presentation using AJAX.
	C310.5	K4 Design and implement simple web page in PHP, and to present data in XML format
	C310.6	K5 Develop the Rich Internet Applications
CS8691- Artificial Intelligence		
45	C311.1	K2 Understand the various characteristics of Intelligent agents
	C311.2	K3 Understand and use appropriate searching algorithms for any AI problem
	C311.3	K3 Learn the problem representation using first order and predicate logic
	C311.4	K3 Understand and design appropriate software agents to solve a problem
	C311.5	K4 Learn the various applications for MLP that use Artificial Intelligence
	C311.6	K5 Understand the role of AI in Healthcare system

CS8601 - Mobile Computing		
46	C312.1	K2 Understand the basics of mobile telecommunication systems
	C312.2	K2 Illustrate the generations of telecommunication systems in wireless networks
	C312.3	K2 Learn and understand the functionality of MAC, network layer and identify a routing protocol for a given Ad hoc network
	C312.4	K3 Understand the functionality of Transport and Application layers
	C312.5	K4 Develop a mobile application using android/blackberry/jos/Windows SDC
	C312.6	K3 Understand the basics of 5G technologies
CS8602 - Compiler Design		
47	C313.1	K2 Understand the different phases of compiler and design a lexical analyzer for a sample language.
	C313.2	K2 Apply different parsing algorithms to develop the parsers for a given grammar
	C313.3	K2 Understand syntax-directed translation
	C313.4	K3 Understand run-time environment and a simple code generator
	C313.5	K3 Learn to implement code optimization techniques
	C313.6	K4 Design and implement a scanner and a parser using LEX and YACC tools
CS8603 - Distributed Systems		
48	C314.1	K2 Elucidate the foundations and issues of distributed systems
	C314.2	K2 Understand the various synchronization issues and global state for distributed systems
	C314.3	K2 Understand the Mutual Exclusion and Deadlock detection algorithms in distributed systems
	C314.4	K3 Describe the agreement protocols and fault tolerance mechanisms in distributed systems
	C314.5	K4 Describe the features of peer-to-peer and distributed shared memory systems
	C314.6	K2 Understand basics of cloud computing
CS8075-Data Warehousing And Data Mining		
49	C315.1	K2 Understand data warehouse concepts, architecture, business analysis with OLAP tools
	C315.2	K2 Learn and apply suitable pre-processing and visualization techniques for data analysis
	C315.3	K3 Study and apply frequent pattern and association rule mining techniques for data analysis
	C315.4	K3 Understand and apply appropriate classification and clustering techniques for data analysis
	C315.5	K4 Learn and understand the use of WEKA tool for the database analysis
	C315.6	K2 Read and understand Sentiment analysis (opinion mining) concepts
IT8076-Software Testing		
50	C316.1	K2 Design test cases suitable for a software development for different domains
	C316.2	K2 Identify suitable tests to be carried out
	C316.3	K3 Prepare test planning based on the document
	C316.4	K3 Document test plans and test cases designed
	C316.5	K4 Use automatic testing tools
	C316.6	K3 Develop and validate a test plan
CS8662- Mobile Application Development Lab		
	C317.1	K2 Develop mobile applications using GUI and Layouts

51	C317.2	K2	Develop mobile applications using Event Listener.
	C317.3	K3	Develop mobile applications using Databases.
	C317.4	K3	Develop mobile applications using KSS Feed, Internal/External Storage, SMS, Multithreading and GPS.
	C317.5	K4	Analyze and discover own mobile app for simple needs.
	C317.6	K4	Develop a mobile application to set wallpaper using Bitmap Class.
CS 8661-Internet Programming Lab			
	C318.1	K2	Able to design Web pages using HTML/XML and style sheets
	C318.2	K3	Able to Create user interfaces using Java frames and applets
	C318.3	K3	Able to Create dynamic web pages using server side scripting
	C318.4	K3	Able to create Client Server applications
	C318.5	K4	Able to Create dynamic web pages using PHP programming
52	C318.6	K5	Able to Create applications with AJAX
HS8581 – Professional Communication			
	C319.1	K1	Take international examination such as IELTS and TOEFL
	C319.2	K2	Make presentations and Participate in Group Discussions
	C319.3	K3	Successfully answers questions in interviews
	C319.4	K3	Effectively communicate inside and outside the classroom.
	C319.5	K4	Make them a flawless fearless proficient speaker and understand engineering ethics and to solve real time problems.
53	C319.6	K4	Acquire sound knowledge in language and linguistics skills (LSRW)
CS8611-Mini Project			
	C320.1	K3	Identify and Formulate a real world challenging problem ethically.
	C320.2	K3	Extract the requirements of the project and analyze extensively the literature survey.
	C320.3	K3	Express technical ideas, strategies and methodologies in written and oral presentations
	C320.4	K4	Learn new tools, algorithms, and/or techniques that contribute to the solution of the project.
	C320.5	K5	Develop sustainable solutions by formulating proper methodology and implement them
54	C320.6	K4	Work as a responsible member and possibly a leader of a team in developing solutions.
Year:2022-2023(ODD) SEMESTER VII			
MG8591-Principles Of Management			
	C401.1	K1	Understanding of management evolution and different types of business.
	C401.2	K2	Knowledge on planning and decision making process of management
	C401.3	K2	Understanding of organization structure, HR planning and control.
	C401.4	K2	Acquaintance of the various process and elements of directing function of management like motivation, leadership and communication.
	C401.5	K3	Designing of performance controlling process, techniques of control and reporting to the management
55	C401.6	K3	To apply principles of management in order to execute the role as a manager in IT industry
CS8792 – Cryptography And Network Security			
	C402.1	K2	Understand the fundamentals of networks security, security architecture, threats and Vulnerabilities

56	C402.2	K2	Gain knowledge on block and stream cipher models and the principles of Public Key Cryptography
	C402.3	K3	Understand the different hash functions and digital signature
	C402.4	K4	Acquire knowledge about Security Practice and System Security
	C402.5	K4	Gain knowledge about Email, IP and Web Security
	C402.6	K5	Know about cyber security
CS8791-Cloud Computing			
57	C403.1	K2	Articulate the main concepts, key technologies, strengths and limitations of cloud computing.
	C403.2	K2	Learn the key and enabling technologies that help in the development of cloud
	C403.3	K3	Develop the ability to understand and use the architecture of compute and storage cloud service and delivery models
	C403.4	K3	Explain the core issues of cloud computing such as resource management and security
	C403.5	K4	Be able to install and use current cloud technologies
58	C403.6	K4	Evaluate and choose the appropriate technologies, algorithms and approaches for implementation and use of cloud
OME752-Supply Chain Management			
58	C404.1	K2	Acquire the scope and importance of supply chain management.
	C404.2	K2	Understand the role of distribution and design distribution networks in supply chain
	C404.3	K3	Elucidate the foundational role of logistics as it relates to transportation and warehousing.
	C404.4	K3	Comprehend the role of sourcing and coordination among suppliers
	C404.5	K4	Develop the supply chain activities using IT framework
59	C404.6	K3	Integrate appropriate technologies in developing solutions to business opportunities and challenges.
OBN752- Hospital Management			
59	C405.1	K2	Gain knowledge about principles of hospital administration
	C405.2	K2	Ability to identify the importance about HR management
	C405.3	K3	Ability to understand recruitment
	C405.4	K3	Ability to understand the information management systems
	C405.5	K4	Ability to understand the safety procedures followed in hospitals
60	C405.6	K2	Ability to analyze the healthcare system
HN8001- Professional Readiness for Innovation, Employability and Entrepreneurship			
60	C406.1	K2	Upskill in emerging technologies and apply to real industry-level use cases
	C406.2	K2	Understand agile development process
	C406.3	K3	Develop career readiness competencies, Team Skills/Leadership qualities
	C406.4	K3	Develop Time management, Project management skills and communication skills
	C406.5	K5	Use critical thinking for innovative problem solving
60	C406.6	K3	Develop entrepreneurship skills to independently work on products.
CS8073- C# and .Net Programming			
60	C407.1	K2	To Write various applications using C# language in the .Net Framework
	C407.2	K2	To Write various applications using C# advanced features in .Net Framework
	C407.3	K3	To Implement base class libraries and database connectivity using ADO .Net

6.1	C407.4	K4	To Develop distributed applications using Java Frameworks
	C407.5	K3	To Implement basic Java libraries and manipulation of data using RMI
	C407.6	K3	Ability to develop program using web services on Java
C48079: Human Computer Interaction			
6.2	C408.1	K2	To Learn the effective dialing for HCI
	C408.2	K3	To Design effective HCI for individuals and persons with disabilities
	C408.3	K3	To Understand the importance of user feedback
	C408.4	K3	To Comprehend the HCI principles for designing multimedia/e-commerce/e-learning to its users
	C408.5	K4	To Develop meaningful user interface
	C408.6	K4	To Understand modern tools for mobile design
C48711: Cloud Computing Lab			
6.3	C409.1	K2	Configure various virtualization tools such as Virtual Box, VMware workstation
	C409.2	K3	Design and deploy a web application in a PaaS environment
	C409.3	K2	Learn how to simulate a cloud environment to implement new schedulers
	C409.4	K3	Install and use a generic cloud environment that can be used as a private cloud
	C409.5	K3	Manipulate large data sets in a parallel environment
	C409.6	K3	Install and usage of open nebula
EW761: Security Lab			
6.4	C410.1	K2	To implement the classical cyber techniques
	C410.2	K3	To implement the modern cyber techniques
	C410.3	K3	To implement the various hash algorithms
	C410.4	K3	To implement the digital signature algorithm
	C410.5	K4	To use different open source tools for implementing network security
	C410.6	K4	To use different open source tools for network intrusion detection
Year 2022-2024 (EVEN) SEMESTER VIII			
C48076: Professional Ethics In Engineering			
6.5	C411.1	K1	Ability to understand the values and ethics
	C411.2	K2	Ability to understand the moral issues
	C411.3	K2	Ability to understand the codes of ethics
	C411.4	K3	Ability to understand the concept of safety and rights
	C411.5	K1	Ability to understand the moral leadership
	C411.6	K3	Ability to understand the environmental and industrial issues
C48079: Green Computing			
	C412.1	K2	To learn the fundamentals of Green Computing
	C412.2	K2	To understand the Green Aspects of enterprise and modelling
	C412.3	K2	To analyze the Green computing tool Framework for virtualization of IT systems
	C412.4	K3	To understand the issues related with Green compliance for Socio-cultural aspects of Green IT

66	C412.5	K4	To study and develop various case studies for The Environmentally Responsible Business Strategies (ERBS)
	C412.6	K3	To apply various modern green computing tools and lead the road on Green Computing
CS8080-Information Retrieval Techniques			
67	C413.1	K1	To learn open source search engine framework and explore its capabilities
	C413.2	K2	Compare and contrast various Information Retrieval models.
	C413.3	K3	Apply appropriate method of clustering or classification
	C413.4	K3	Design and implement innovative features in a search engine
	C413.5	K4	To design and implement recommender system
	C413.6	K3	To learn deep neural network
CS8811-Project Work			
68	C414.1	K3	Formulate a real world challenging problem ethically
	C414.2	K3	Develop the requirements of the project and analyse extensively the literature survey.
	C414.3	K3	Express technical ideas, strategies and methodologies in written and oral presentations.
	C414.4	K5	Develop sustainable solutions by formulating proper methodology and implement them.
	C414.5	K4	Self-learn new tools, algorithms, and/or techniques that contribute to the software solution of the project
	C414.6	K3	Work as a responsible member and possibly a leader of a team in developing software solutions.

S.A ENGINEERING COLLEGE, CHENNAI - 77
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COURSE OUTCOMES
2018-2022 BATCH

Sl.No	COURSE NAME	LEVEL MAPPING	COURSE OUTCOMES
Year:2018-2019(ODD) SEMESTER I			
HS8151 - Communicative English			
1	C101.1	K1	Participate effectively in informal conversation, introduce themselves and their friends and express opinions in English
	C101.2	K1	Comprehend conversations and short talks delivered in English.
	C101.3	K2	Read and write in a coherent and organized manner.
	C101.4	K2	Read articles of a general kind in and newspapers
	C101.5	K2	Write short essays of a general kind and personal letters and emails in English.
	C101.6	K2	Can communicate any idea of a general kind in an organized manner.
MA8151 - Engineering Mathematics - I			
2	C102.1	K1	Learn the basic concepts to use both the limit definition and rules of differentiation to differentiate functions.
	C102.2	K1	Apply partial differentiation to solve maxima and minima problems.
	C102.3	K2	Evaluate integrals both by using Riemann sums and by using the Fundamental Theorem of Calculus.
		K2	Evaluate Integrals using techniques of integration, such as substitution, partial fractions and integration by parts
	C102.4	K1	Determine convergence/divergence of improper integrals and evaluate convergent improper integrals
	C102.5	K2	Apply integration to compute multiple integrals, area, volume, integrals in polar coordinates, in addition to change of order and change of variables
	C102.6	K1	Apply various techniques in solving differential equations.
PH8151 - Engineering Physics			
3	C103.1	K1	Gain knowledge on the basics of properties of matter and its applications
	C103.2	K1	Acquire knowledge on the concepts of waves and optical devices and their applications in fiber optics
		K2	Get Adequate knowledge on the concepts of thermal properties of materials and their applications in expansion joints and heat exchangers
	C103.3	K2	Get Adequate knowledge on the concepts of thermal properties of materials and their applications in expansion joints and heat exchangers
	C103.4	K1	Get knowledge on advanced physics concepts of quantum theory and its applications in tunneling microscopes
	C103.5	K2	Understand the basics of crystals, their structures and different crystal growth techniques
	C103.6	K1	Acquire knowledge on solid state laser, graphite and zinc blende structures.
CY8151 - Engineering Chemistry			
4	C104.1	K1	About the type of troubles faced in boilers and the methods used to treat impure water.
	C104.2	K1	The factors affecting the rate of adsorption and catalytic activity.
	C104.3	K1	With the help of phase rule, they could understand the various phase diagram and able to predict the low melting alloys
		K1	Types of solid, liquid and gaseous fuel, manufacturing methods and basic steps involved in combustion reaction
	C104.5	K1	Types of batteries, their reactions and the importance of renewable energy resources.
	C104.6	K2	The basics and applications of nanomaterial in the recent innovations.
GE8151 - Problem Solving and Python Programming			
5	C105.1	K1	Understand the basic of algorithmic problem solving
	C105.2	K1	Be familiar with data expressions and statements
	C105.3	K2	understand control flow and functions problems
	C105.4	K1	comprehend list, tuples and dictionaries.
	C105.5	K1	Read and write data from/to files in Python Programs
	C105.6	K2	Understand Object oriented programming concepts
GE8152 - Engineering Graphics			
6	C106.1	K1	Understand existing national standards of engineering drawing and visualization concepts
	C106.2	K1	Understand the projection of Points, Lines and Plane surfaces
	C106.3	K1	Understand the projection of Solids
	C106.4	K1	Understand the section of solids
	C106.5	K1	Understand the development of surfaces
	C106.6	K1	Understand the isometric and perspective projections
GE8161 - Problem Solving and Python Programming Laboratory			
7	C107.1	K2	Write, test, and debug simple Python programs.
	C107.2	K3	Implement Python programs with conditionals and loops.
	C107.3	K3	Develop Python programs step-wise by defining functions and calling them.
	C107.4	K2	Use Python lists, tuples, dictionaries for representing compound data.
	C107.5	K3	Read and write data from/to files in Python
	C107.6	K2	Design and implement a program to solve a real-world problem using the language idioms, data structure and standard library
BS8161 - Physics and Chemistry Laboratory			
	C108.1	K1	Apply principles of optics, sound and thermal properties for engineering.
	C108.2	K2	Apply principles of elasticity for engineering applications.
	C108.3	K3	Analyse the viscosity of a liquid.

8	C108.4	K2	Apply hands-on knowledge in the quantitative chemical analysis of water.
	C108.5	K1	Carryout the basics of instrumental analysis-conductivity meter, potentiometer, pH meter
	C108.6	K2	Analyse the total dissolved solids in water
Year:2018-2019(EVEN) SEMESTER II			
HS8251- Technical English			
9	C109.1	K1	Learning the concepts of applying differential equation in signals
	C109.2	K2	Listen and comprehend lectures and talks in their area of specialization successfully.
	C109.3	K2	Speak appropriately and effectively in varied formal and informal contexts
	C109.4	K2	Write reports and writing job applications
	C109.5	K2	Write minutes of meeting and to take part in group discussion effectively.
	C109.6	K2	Attain fluency in English language to speak convincingly, express their opinions clearly, initiate a discussion, negotiate and comprehend using appropriate communicative strategies.
MA8251 - Engineering Mathematics - II			
10	C110.1	K1	Eigen values and Eigen vectors, diagonalization of a matrix, Symmetric matrices, Positive definite matrices and similar matrices.
	C110.2	K2	Gradient, Divergence and curl of a vector field and related identities, evaluation of line, surface and volume integrals using Gauss, Stokes' and Green's theorems and their verification.
	C110.3	K2	Analytic functions, conformal mapping
	C110.4	K3	Complex Integration
	C110.5	K3	Laplace transform and inverse transform of simple functions, properties, various related theorems and application to differential equations with constant coefficients
	C110.6	K4	Learning the concepts of applying differential equation in signals
PH8252 - Physics for Information Science			
11	C111.1	K1	To gain knowledge on industrial application of DC motors
	C111.2	K1	Acquire knowledge on basics of semiconductor physics and its applications in various devices
	C111.3	K2	Get knowledge on magnetic properties of materials and their applications in data storage
	C111.4	K2	Have the necessary understanding on the functioning of optical materials for optoelectronics
	C111.5	K2	Understand the basics of quantum structures and their applications in carbon electronics.
	C111.6	K3	Gain knowledge on application of semiconducting materials and preparation of nano materials.
BE8255 - Basic Electrical, Electronics and Measurement Engineering			
12	C112.1	K1	Understand and analyse the fundamental laws & electrical network theorems
	C112.2	K1	Gain knowledge on DC & AC static and rotating machines
	C112.3	K1	Gain knowledge on renewable energy sources, various electrical protective devices
	C112.4	K2	To understand the fundamentals of electronic circuits
	C112.5	K2	To gain knowledge on the working principle & operation of measuring instruments and transducers.
	C112.6	K2	To gain knowledge on industrial application of DC motors
GE8291 - Environmental Science and Engineering			
13	C113.1	K1	Students will be able to understand the functions of ecosystems and appreciate the bio diversity.
	C113.2	K2	Students will be able to know the measures to control environmental pollution.
	C113.3	K1	Students will be able to understand the usage as well as the effects of over exploitation of natural resources.
	C113.4	K1	Students will have knowledge about finding technological, economic and political solutions to environmental problems with various Environmental Protection Act in mind.
	C113.5	K2	Students will be able to understand the interrelationship between population explosion and the environment and also role of IT in environment and human health.
	C113.6	K3	Students will be able to understand that Environmental problems can only be solved by Public participation in all aspects and cannot be solved by mere laws.
CS8251 - Programming in C			
14	C114.1	K2	Develop simple applications in C using basic constructs
	C114.2	K2	Design and implement applications using arrays and strings
	C114.3	K3	Develop and implement applications in C using functions and pointers.
	C114.4	K3	Develop applications in C using structures.
	C114.5	K3	Design applications using sequential and random access file processing.
	C114.6	K4	Understand the concept of dynamic memory management and basic data structures such as arrays and linked list
GE8261 - Engineering Practices Laboratory			
15	C115.1	K1	Understand the basic principle of welding
	C115.2	K2	Understand the basic principle of Sheet Metal Work
	C115.3	K2	Understand the basic principle of joints
	C115.4	K2	Understand the basic principle of Machine assembly practice
	C115.5	K2	Understand the basic principle of Drilling
	C115.6	K3	Understand the basic principle of turning
CS8261- C Programming Laboratory			
	C116.1	K2	Read, understand and trace the execution of programs written in C language.

16	C116.2	K3	Develop conditional and iterative statements to write C programs
	C116.3	K3	Develop C programs for simple applications making use of basic constructs, arrays and strings.
	C116.4	K3	Develop C programs involving functions, recursion, pointers, and structures
	C116.5	K2	Design applications using sequential and random access file processing.
	C116.6	K4	Exercise user defined functions to solve real time problems
Year:2019-2020(ODD) SEMESTER III			
MA8351 - Discrete Mathematics			
17	C201.1	K1	Have knowledge of the concepts needed to test the logic of a program.
	C201.2	K2	Be aware of the counting principles
	C201.3	K2	Learn the concepts of Graph Theory that would help them to define new levels of networks which are implemented in AI and ANN.
	C201.4	K2	Be exposed to concepts and properties of algebraic structures such as groups, rings and fields.
	C201.5	K3	Expose the concepts and properties of Lattices and Boolean algebra used in Coding and Decoding theory of Cryptography.
	C201.6	K4	most of the basic terminologies used in computer science courses and application of ideas to solve practical problems.
CS8351 - Digital Principles and System Design			
18	C202.1	K1	Simplify Boolean functions using K-Map
	C202.2	K2	Design and Analyze Combinational Circuits
	C202.3	K2	Design and Analyze Sequential Circuits.
	C202.4	K2	Implement designs using Programmable Logic Devices
	C202.5	K3	Write HDL code for combinational and Sequential Circuits
	C202.6	K4	ASM chart for digital circuits
CS8391 - Data Structures			
19	C203.1	K2	Learn and apply the concept and operations of List ADT
	C203.2	K2	Understand and apply the concept and operations of Stack and Queue ADT
	C203.3	K2	Gain the knowledge about Tree ADT and its Applications
	C203.4	K3	Understand the concept, operations and applications of Graph ADT
	C203.5	K3	Understand and analyze sorting, searching and hashing Techniques
	C203.6	K4	Comprehend Minimum Spanning Trees – and algorithms (Kruskal and Prims algorithm)
CS8392 - Object Oriented Programming			
20	C204.1	K2	Understand the fundamentals of OOP using Java.
	C204.2	K2	Learn the concepts of inheritance, interfaces
	C204.3	K2	Build Java applications using exceptions and I/O streams
	C204.4	K3	Develop Java applications with threads and generics classes
	C204.5	K3	Develop Interactive Java programs using swings.
	C204.6	K4	Learn the concepts of Applets and develop the interactive Java programs using Applets
EC8395 - Communication Engineering			
21	C205.1	K2	Learn the concepts of Applets and develop the interactive Java programs using Applets
	C205.2	K2	Apply analog and digital communication techniques.
	C205.3	K2	Use data and pulse Communication techniques.
	C205.4	K3	Analyze Source and Error control coding
	C205.5	K3	Apply spread spectrum modulation techniques
	C205.6	K4	Analyze basic concepts of cellular communication.
CS8381 - Data Structures Laboratory			
22	C206.1	K2	Implement the operations of List, Stack and Queue using Array
	C206.2	K2	Implement the operations of List, Stack and Queue using Linked List
	C206.3	K2	Understand and implement the different operations of various Trees
	C206.4	K3	Implement graph traversal algorithms and techniques
	C206.5	K3	Understand and implement various sorting, searching and hashing algorithms
	C206.6	K4	Implement Minimum Spanning Trees algorithm
CS8383 - Object Oriented Programming Laboratory			
23	C207.1	K2	Develop and Implement Java programs for simple applications that make use of classes, packages and Interfaces.
	C207.2	K2	Develop and Implement Java programs with array list and exception handling
	C207.3	K2	Develop and Implement Java programs using the concept of Multithreading.
	C207.4	K3	Design and develop the applications using file processing, generic programming
	C207.5	K3	Design and develop the applications using event handling mechanism
	C207.6	K4	Develop the mini-project using the concepts of Java Programming
CS8382 - Digital Systems Laboratory			
24	C208.1	K1	Implement simplified combinational circuits using basic logic gates
	C208.2	K2	Implement combinational circuits using MSI devices
	C208.3	K2	Implement sequential circuits like registers and counters
	C208.4	K3	Simulate combinational circuits using HDL
	C208.5	K3	Simulate sequential circuits using HDL
	C208.6	K4	Implement various digital circuits
HS8381 - Interpersonal Skills/Listening & Speaking			
	C209.1	K1	Listen and respond appropriately.
	C209.2	K2	Participate in group discussions
	C209.3	K2	Make effective presentations
	C209.4	K2	Participate confidently and appropriately in conversations both formal and informal
	C209.5	K3	Persuade through conversations

25	C209.0	K4	Improve pronunciation clearly.
Year:2019-2020(EVEN) SEMESTER IV			
MA0402 - Probability and Queueing Theory			
26	C210.1	K1	Understand the fundamental knowledge of the concepts of probability and have knowledge of standard distributions which can describe real life phenomenon.
	C210.2	K2	Understand the basic concepts of one and two dimensional random variables and apply in engineering applications.
	C210.3	K3	Apply the concept of random processes in engineering disciplines.
	C210.4	K3	Acquire skills in analyzing queueing models.
	C210.5	K2	Understand and characterize phenomenon which evolve with respect to time in a probabilistic manner.
26	C210.6	K3	Learn the required mathematical support in real life problems and develop probabilistic models which can be used in several areas of science and engineering.
CS8491 - Computer Architecture			
27	C211.1	K2	Understand the basic structure of computers, operations and instructions.
	C211.2	K2	Ability to Design arithmetic and logic unit.
	C211.3	K2	Understand pipelined execution and design control unit.
	C211.4	K3	Understand parallel processing architectures.
	C211.5	K4	Understand the various memory systems and I/O communication.
	C211.6	K3	Understand Homogeneous and heterogeneous processors.
CS8492 - Database Management Systems			
28	C212.1	K2	Classify the modern and futuristic database applications based on size and complexity.
	C212.2	K3	Map ER model to Relational model to perform database design effectively.
	C212.3	K2	Write queries using normalization criteria and optimize queries.
	C212.4	K3	Compare and contrast various indexing strategies in different database systems.
	C212.5	K4	Appraise how advanced databases differ from traditional databases.
	C212.6	K4	Study about various open source databases.
CS8451 - Design and Analysis of Algorithms			
29	C213.1	K2	Analyze the time and space complexity of algorithms.
	C213.2	K2	Design algorithms for various computing problems (brute-force, divide and conquer).
	C213.3	K3	Analyze the different algorithm design techniques for a given problem.
	C213.4	K2	Understand the application of iterative algorithm.
	C213.5	K3	Modify existing algorithms to improve efficiency.
	C213.6	K4	Adopt the recent algorithm design technique to solve the given problem.
CS8493 - Operating Systems			
30	C214.1	K2	Prepare reports for real life database applications.
	C214.2	K2	Understand the principles of concurrency, design various CPU scheduling algorithms and Deadlock mechanisms.
	C214.3	K3	Compare and contrast various memory management schemes.
	C214.4	K4	Understand the prototype for file systems.
	C214.5	K3	Perform administrative tasks on Linux Servers and compare iOS and Android OS.
	C214.6	K4	Acquire knowledge about Operating systems used for Business applications.
CS8494 - Software Engineering			
31	C215.1	K2	Identify key activities in software process and compare different process models.
	C215.2	K2	Understand the concepts of requirement engineering and analysis modeling.
	C215.3	K4	Applying the systematic procedure for software design.
	C215.4	K3	Compare and Contrast various testing methods and implementation techniques.
	C215.5	K3	Management techniques used for project schedule, estimation of project cost.
	C215.6	K4	Understand the evolution of object oriented models.
CS8481 - Database Management Systems Laboratory			
32	C216.1	K2	Use typical data definitions and manipulation commands.
	C216.2	K2	Design applications to test Nested and Join Queries.
	C216.3	K3	Implement simple applications that use Views, Sequences, Synonyms and Cursors.
	C216.4	K3	Design database using ER modeling, Normalize and Implement applications that require a Front-end Tool.
	C216.5	K3	Critically analyze the use of Tables, Functions, Procedures, Triggers and Exception Handling using PL/SQL.
	C216.6	K4	Prepare reports for real life database applications.
CS8461 - Operating Systems Laboratory			
33	C217.1	K2	Compare the performance of various CPU Scheduling Algorithms.
	C217.2	K2	Implement Deadlock avoidance and Detection Algorithms.
	C217.3	K2	Implement Semaphores.
	C217.4	K3	Create processes and implement IPC.
	C217.5	K3	Analyze the performance of the various Page Replacement Algorithms.
	C217.6	K4	Implement File Organization and File Allocation Strategies.
HS8461 - Advanced Reading and Writing			
34	C218.1	K1	Write different types of essays.
	C218.2	K1	Write winning job applications.
	C218.3	K2	Read and evaluate texts critically.
	C218.4	K2	Display critical thinking in various professional contexts.
	C218.5	K3	Communicate effectively through mails.
	C218.6	K2	Justify the purpose of higher studies.
Year:2020-2021(ODD) SEMESTER V			
MA8551 - Algebra and Number Theory			

35	C301.1	K1	Apply the basic notions of groups, rings, fields which will then be used to solve related problems
	C301.2	K2	Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts
	C301.3	K2	Demonstrate accurate and efficient use of advanced algebraic techniques
	C301.4	K2	Demonstrate their mastery by solving non – trivial problems related to the concepts, and by proving simple theorems about the statements proven by the text.
	C301.5	K3	Apply integrated approach to number theory and abstract algebra, and provide a firm basis for further reading and study in the subject.
	C301.6	K3	An ability to deal with abstraction and to introduce most of the basic concepts in number theory used in computer science courses and application in cryptography and network security
CS8591 - Computer Networks			
36	C302.1	K2	Understand the basic layers and its functions in computer networks
	C302.2	K2	Understand the basics of how data flows from one node to another
	C302.3	K3	Learn the functions of network layer and the various routing protocols
	C302.4	K3	Understand the functions and protocols of Transport layer
	C302.5	K3	Understand the working of various application layer protocols.
	C302.6	K3	Gain ideas on security issues of networks
EC8691 - Microprocessors and Microcontrollers			
37	C303.1	K2	Understand the Architecture of 8086 microprocessor and ALP
	C303.2	K2	Design the bus architecture and multiprocessor configuration of 8086 microprocessor
	C303.3	K3	Design the I/O and Memory Interfacing circuits using 8086 and relevant case studies
	C303.4	K2	Understand the Architecture of 8051 microcontroller and ALP
	C303.5	K4	Design the various interfacing circuits using 8051
	C303.6	K3	Comprehend the Architecture of modern microprocessor
CS8501 - Theory of Computation			
38	C304.1	K2	Construct automata for any pattern
	C304.2	K2	Construct regular expression for any pattern
	C304.3	K3	Write Context free grammar for any construct
	C304.4	K4	Design Turing machines for any language and propose computation solutions
	C304.5	K3	Derive whether a problem is decidable or not.
	C304.6	K3	Understand Context Sensitive Grammar and LBA
CS8592 - Object Oriented Analysis and Design			
39	C305.1	K2	Express software design with UML diagrams.
	C305.2	K4	Design software applications using OO concepts
	C305.3	K3	Identify various scenarios based on software requirements.
	C305.4	K3	Transform UML based software design into pattern based design using design patterns.
	C305.5	K2	Understand the various testing methodologies for OO software
	C305.6	K2	Compare and contrast various testing techniques
OCE552- Geographic Information System			
40	C306.1	K1	Understand the basic idea about the fundamentals of GIS
	C306.2	K2	Understand the types spatial of data models
	C306.3	K2	Gain knowledge about data input and topology
	C306.4	K2	Gain knowledge on data analysis tools and network analysis.
	C306.5	K3	Understand data management functions and data output.
	C306.6	K2	Understand the GIS in the Cloud
EC8681 - Microprocessors and Microcontrollers Laboratory			
41	C307.1	K2	Develop ALP Programs for fixed and Floating Point and Arithmetic using 8086
	C307.2	K3	Interface different I/Os with 8086 processor
	C307.3	K2	Generate waveforms using 8086 Microprocessors
	C307.4	K3	Develop and execute programs using 8051 Microcontroller
	C307.5	K3	Develop and execute programs using MASM.
	C307.6	K2	Study and Develop ALP program using ARM processor
CS8582 - Object Oriented Analysis and Design Laboratory			
42	C308.1	K2	Perform OO analysis and design for a given problem specification.
	C308.2	K2	Identify and map basic software requirements in UML mapping
	C308.3	K3	Improve the software quality using design patterns and to explain the rationale behind applying specific design patterns
	C308.4	K3	Test the compliance of the software with the SRS
	C308.5	K3	Generate code from design.
	C308.6	K4	Design and develop a case study on object oriented projects
CS8581 - Networks Laboratory			
43	C309.1	K2	Implement various protocols using TCP and UDP.
	C309.2	K2	Compare the performance of different transport layer protocols.
	C309.3	K3	Use simulation tools to analyze the performance of various network protocols.
	C309.4	K4	Analyze various routing algorithms
	C309.5	K4	Implement error correction codes.
	C309.6	K2	Study about the configuration of Cisco Network Operating system.
Year:2020-2021(EVEN) SEMESTER VI			
CS8651 - Internet Programming			
	C310.1	K2	Understand the java programming.
	C310.2	K2	Able to design a basic website using HTML and Cascading Style Sheets.

44	C310.3	K3	Able to Design and implement dynamic web page with validation using JavaScript objects and by applying different event handling mechanisms.
	C310.4	K3	Able to Design rich client presentation using AJAX.
	C310.5	K4	Design and implement simple web page in PHP, and to present data in XML format.
	C310.6	K5	Develop the Rich Internet Applications.
CS8691- Artificial Intelligence			
45	C311.1	K2	Understand the various characteristics of intelligent agents.
	C311.2	K3	Understand and use appropriate searching algorithms for any AI problem.
	C311.3	K3	Learn the problem representation using first order and predicate logic.
	C311.4	K3	Understand and design appropriate software agents to solve a problem.
	C311.5	K4	Learn the various applications for NLP that use Artificial Intelligence.
	C311.6	K5	Understand the role of AI in Healthcare system.
CS8601 - Mobile Computing			
46	C312.1	K2	Understand the basics of mobile telecommunication systems.
	C312.2	K2	Illustrate the generations of telecommunication systems in wireless networks.
	C312.3	K2	Learn and understand the functionality of MAC, network layer and identify a routing protocol for a given Ad hoc network.
	C312.4	K3	Understand the functionality of Transport and Application layers.
	C312.5	K4	Develop a mobile application using android/blackberry/ios/Windows SDK.
	C312.6	K3	Understand the basics of 5G technologies.
CS8602 - Compiler Design			
47	C313.1	K2	Understand the different phases of compiler and design a lexical analyzer for a sample language.
	C313.2	K2	Apply different parsing algorithms to develop the parsers for a given grammar.
	C313.3	K2	Understand syntax-directed translation.
	C313.4	K3	Understand run-time environment and a simple code generator.
	C313.5	K3	Learn to implement code optimization techniques.
	C313.6	K4	Design and implement a scanner and a parser using LEX and YACC tools.
CS8603 - Distributed Systems			
48	C314.1	K2	Elucidate the foundations and issues of distributed systems.
	C314.2	K2	Understand the various synchronization issues and global state for distributed systems.
	C314.3	K2	Understand the Mutual Exclusion and Deadlock detection algorithms in distributed systems.
	C314.4	K3	Describe the agreement protocols and fault tolerance mechanisms in distributed systems.
	C314.5	K4	Describe the features of peer-to-peer and distributed shared memory systems.
	C314.6	K2	Understand basics of cloud computing.
CS8076-Data Warehousing And Data Mining			
49	C315.1	K2	Understand data warehouse concepts, architecture, business analysis with OLAP tools.
	C315.2	K2	Learn and apply suitable pre-processing and visualization techniques for data analysis.
	C315.3	K3	Study and apply frequent pattern and association rule mining techniques for data analysis.
	C315.4	K3	Understand and apply appropriate classification and clustering techniques for data analysis.
	C315.5	K4	Learn and understand the use of WEKA tool for the database analysis.
	C315.6	K2	Read and understand Sentiment analysis (opinion mining) concepts.
IT8076-Software Testing			
50	C316.1	K2	Design test cases suitable for a software development for different domains.
	C316.2	K2	Identify suitable tests to be carried out.
	C316.3	K3	Prepare test planning based on the document.
	C316.4	K3	Document test plans and test cases designed.
	C316.5	K4	Use automatic testing tools.
	C316.6	K3	Develop and validate a test plan.
CS8662- Mobile Application Development Lab			
51	C317.1	K2	Develop mobile applications using GUI and Layouts.
	C317.2	K2	Develop mobile applications using Event Listener.
	C317.3	K3	Develop mobile applications using Databases.
	C317.4	K3	Develop mobile applications using RSS Feed, Internal/External Storage, SMS, Multithreading and GPS.
	C317.5	K4	Analyze and discover own mobile app for simple needs.
	C317.6	K4	Develop a mobile application to set wallpaper using Bitmap Class.
CS 8661-Internet Programming Lab			
52	C318.1	K2	Able to design Web pages using HTML/XML and style sheets.
	C318.2	K3	Able to Create user interfaces using Java frames and applets.
	C318.3	K3	Able to Create dynamic web pages using server side scripting.
	C318.4	K3	Able to create Client Server applications.
	C318.5	K4	Able to Create dynamic web pages using PHP programming.
	C318.6	K5	Able to Create applications with AJAX.
HS8581 - Professional Communication			
	C319.1	K1	Take international examination such as IELTS and TOEFL.
	C319.2	K2	Make presentations and Participate in Group Discussions.
	C319.3	K3	Successfully answers questions in interviews.
	C319.4	K3	Effectively communicate inside and outside the classroom.

53	C310.5	K4	Make them a flawless fearless proficient speaker and understand engineering ethics and to solve real time problems.
	C310.6	K4	Acquire sound knowledge in language and linguistics skills (LSRW)
CS8611-Mini Project			
54	C320.1	K3	Identify and Formulate a real world challenging problem ethically
	C320.2	K3	Extract the requirements of the project and analyze extensively the literature survey.
	C320.3	K3	Express technical ideas, strategies and methodologies in written and oral presentations
	C320.4	K4	Learn new tools, algorithms, and/or techniques that contribute to the solution of the project
	C320.5	K5	Develop sustainable solutions by formulating proper methodology and implement them
54	C320.6	K4	Work as a responsible member and possibly a leader of a team in developing solutions
Year:2021-2022(ODD) SEMESTER VII			
MG8591-Principles Of Management			
55	C401.1	K1	Understanding of management evolution and different types of business.
	C401.2	K2	Knowledge on planning and decision making process of management
	C401.3	K2	Understanding of organization structure, HR planning and control.
	C401.4	K2	Acquaintance of the various process and elements of directing function of management like motivation, leadership and communication.
	C401.5	K3	Designing of performance controlling process, techniques of control and reporting to the management
	C401.6	K3	To apply principles of management in order to execute the role as a manager in IT industry
CS8792 – Cryptography And Network Security			
56	C402.1	K2	Understand the fundamentals of networks security, security architecture, threats and Vulnerabilities
	C402.2	K2	Gain knowledge on block and stream cipher models and the principles of Public Key Cryptography
	C402.3	K3	Understand the different hash functions and digital signature
	C402.4	K4	Acquire knowledge about Security Practice and System Security
	C402.5	K4	Gain knowledge about Email, IP and Web Security
	C402.6	K5	Know about cyber security..
CS8791-Cloud Computing			
57	C403.1	K2	Articulate the main concepts, key technologies, strengths and limitations of cloud computing.
	C403.2	K2	Learn the key and enabling technologies that help in the development of cloud.
	C403.3	K3	Develop the ability to understand and use the architecture of compute and storage cloud, service and delivery models.
	C403.4	K3	Explain the core issues of cloud computing such as resource management and security.
	C403.5	K4	Be able to install and use current cloud technologies.
	C403.6	K4	Evaluate and choose the appropriate technologies, algorithms and approaches for implementation and use of cloud
OME752 -Supply Chain Management			
58	C404.1	K2	Acquire the scope and importance of supply chain management.
	C404.2	K2	Understand the role of distribution and design distribution networks in supply chain.
	C404.3	K3	Elucidate the foundational role of logistics as it relates to transportation and warehousing
	C404.4	K3	Comprehend the role of sourcing and coordination among suppliers.
	C404.5	K4	Develop the supply chain activities using IT framework.
	C404.6	K3	Integrate appropriate technologies in developing solutions to business opportunities and challenges.
CS8091- Big Data Analytics			
59	C405.1	K2	Work with big data tools and its analysis techniques
	C405.2	K2	Analyze data by utilizing clustering and classification algorithms
	C405.3	K3	Learn and apply different mining algorithms and recommendation systems for large volumes of data
	C405.4	K3	Perform analytics on data streams
	C405.5	K4	Learn NoSQL databases and management
	C405.6	K2	To learn and apply data visualization tool like Tableau
IT8075-Software Project Management			
60	C406.1	K2	Understand Project Management principles while developing software.
	C406.2	K2	Obtain adequate knowledge about software process models and software effort estimation techniques
	C406.3	K3	Estimate the risks involved in various project activities.
	C406.4	K3	Define the checkpoints, project reporting structure, project progress and tracking mechanisms using project management principles
	C406.5	K3	Learn staff selection process and the issues related to people management.
	C406.6	K3	Obtain knowledge on managing people in a software development team

CS8073- C# and .Net Programming			
61	C407.1	K2	To Write various applications using C# language in the .Net Framework.
	C407.2	K2	To Write various applications using C# advanced features in .Net Framework.
	C407.3	K3	To Implement base class libraries and database connectivity using ADO.Net.
	C407.4	K4	To Develop distributed applications using .Net Framework.
	C407.5	K3	To Implement base class libraries and manipulation of data using XML.
	C407.6	K3	Ability to develop program using webservice in .Net
CS8079- Human Computer Interaction			
62	C408.1	K2	To Learn the effective dialog for HCI
	C408.2	K3	To Design effective HCI for individuals and persons with disabilities
	C408.3	K3	To Understand the importance of user feedback
	C408.4	K3	To Comprehend the HCI implications for designing multimedia/ ecommerce/ e-learning Web sites
	C408.5	K4	To Develop meaningful user interface
	C408.6	K4	To Understand modern tools for mobile design
CS8711 -Cloud Computing Lab			
63	C409.1	K2	Configure various virtualization tools such as Virtual Box, VMware workstation.
	C409.2	K3	Design and deploy a web application in a PaaS environment.
	C409.3	K2	Learn how to simulate a cloud environment to implement new schedulers.
	C409.4	K3	Install and use a generic cloud environment that can be used as a private cloud.
	C409.5	K3	Manipulate large data sets in a parallel environment.
	C409.6	K3	Install and usage of open nebula
IT8761-Security Lab			
64	C410.1	K2	To implement the classical cipher techniques
	C410.2	K3	To implement the modern cipher techniques
	C410.3	K3	To implement the various hash algorithms
	C410.4	K3	To implement the digital signature algorithm
	C410.5	K4	To use different open source tools for implementing network security
	C410.6	K4	To use different open source tools for network intrusion detection
Year:2021-2022(EVEN) SEMESTER VIII			
GE8076-Professional Ethics In Engineering			
65	C411.1	K1	Ability to understand the values and ethics
	C411.2	K2	Ability to understand the moral issues
	C411.3	K2	Ability to understand the codes of ethics.
	C411.4	K3	Ability to understand the concept of safety and rights
	C411.5	K1	Ability to understand the moral leadership
	C411.6	K3	Ability to understand the environmental and industrial issues
CS8078-Green Computing			
66	C412.1	K2	To learn the fundamentals of Green Computing.
	C412.2	K2	To understand the Green Assets of enterprise and modelling
	C412.3	K2	To analyze the Green computing Grid Framework for virtualization of IT systems.
	C412.4	K3	To understand the issues related with Green compliance for Socio-cultural aspects of Green IT
	C412.5	K4	To study and develop various case studies for The Environmentally Responsible Business Strategies (ERBS)
	C412.6	K3	To apply various modern green computing tools and lead the road on Green Computing
CS8080-Information Retrieval Techniques			
67	C413.1	K1	To learn open source search engine framework and explore its capabilities
	C413.2	K2	Compare and contrast various Information Retrieval models.
	C413.3	K3	Apply appropriate method of clustering or classification
	C413.4	K3	Design and implement innovative features in a search engine.
	C413.5	K4	To design and implement recommender system
	C413.6	K3	To learn deep neural network
CS8811-Project Work			
68	C414.1	K3	Formulate a real world challenging problem ethically.
	C414.2	K3	Develop the requirements of the project and analyse extensively the literature survey.
	C414.3	K3	Express technical ideas, strategies and methodologies in written and oral presentations.
	C414.4	K4	Develop sustainable solutions by formulating proper methodology and implement them.
	C414.5	K4	Self-learn new tools, algorithms, and/or techniques that contribute to the software solution of the project
	C414.6	K5	Work as a responsible member and possibly a leader of a team in developing software solutions.